AF3060 INVERTER DRIVES

DRIVER - 331X351AAG01, G02

OPERATION – TROUBLE SHOOTING – REPAIR

GEK-62501A

DRIVER ASSEMBLY	ELEM	36D865532AA
	CONN	36C764841AA

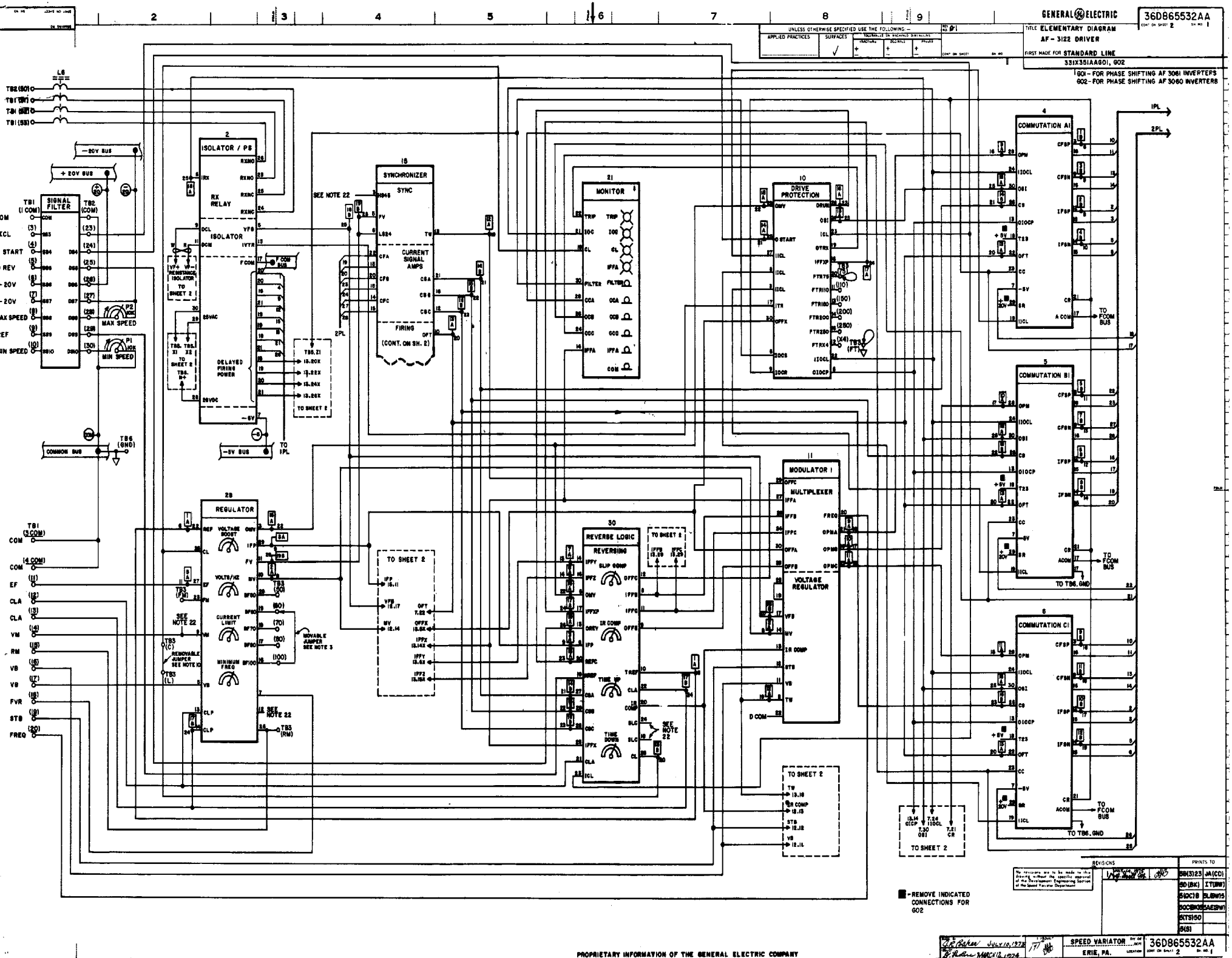
PRINTED CIRCUIT CARD DIAGRAMS

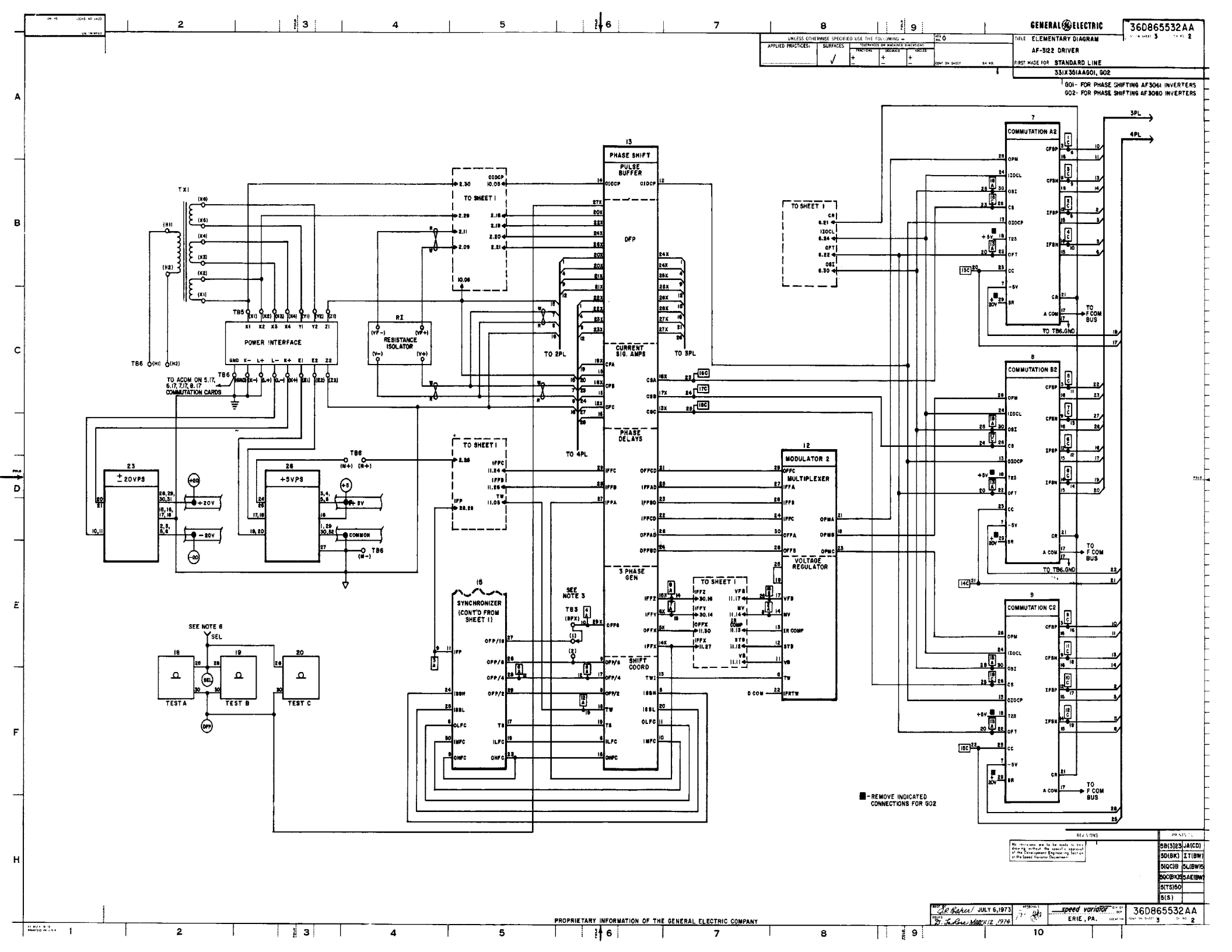
DRIVE PROTECTION	36C762849AC
COMMUTATION	36C764126AB
REGULATOR	36C764136AB
MODULATOR	36C764107AA
SYNCHRONIZER	36C762853AC
REVERSE LOGIC	36C762855AB
SIGNAL FILTER	36C762868AA
MONITOR	36C762854AA
ISOLATOR/POWER SUPPLY	36C762869BR
5V POWER SUPPLY	36C762870AB
20V POWER SUPPLY	36C762896AB
TEST CARD	36C762858AA
POWER INTERFACE	36C764129AA
RESISTANCE ISOLATOR	36C764140AA
PHASE SHIFT	36C762893CA

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company

GENERAL ELECTRIC

demark of the General Electric Co.


GENERAL ELECTRIC
36D865532AA
TITLE ELEMENTARY DIAGRAM
AF-3122 DRIVER
FIRST MADE FOR STANDARD LINE
331X351AA601, 602
601 - FOR PHASE SHIFTING AF 3061 INVERTERS
602 - FOR PHASE SHIFTING AF 3060 INVERTERS
SIGNAL FILTER
ISOLATOR / PS
RX RELAY
ISOLATOR
DELAYED FIRING POWER
SYNCHRONIZER
SYNC
CURRENT SIGNAL AMPS
FIRING
(CONT. ON SH. 2)
MONITOR
DRIVE PROTECTION
REGULATOR
VOLTAGE BOOST
VOLTS/HZ
CURRENT LIMIT
MINIMUM FREQ
REVERSE LOGIC
REVERSING
SLIP COMP
IR COMP
TIME UP
TIME DOWN
MODULATOR 1
MULTIPLEXER
VOLTAGE REGULATOR
COMMUTATION A1
COMMUTATION B1
COMMUTATION C1
MAX SPEED
MIN SPEED
COMMON BUS
-20V BUS
+20V BUS
-5V BUS
TO IPL
TO FCOM BUS
TO TB6, GND
TO SHEET 2
SEE NOTE 22
REMOVABLE JUMPER SEE NOTE 10
MOVABLE JUMPER SEE NOTE 3
■ - REMOVE INDICATED CONNECTIONS FOR 602
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR
ERIE, PA.
36D865532AA


GENERAL ELECTRIC
36D865532AA
TITLE ELEMENTARY DIAGRAM
AF-3122 DRIVER
FIRST MADE FOR STANDARD LINE
331X351AAG01, G02
G01- FOR PHASE SHIFTING AF3061 INVERTERS
G02- FOR PHASE SHIFTING AF3060 INVERTERS
TO SHEET 1
TO 2PL
TO 3PL
TO 4PL
PHASE SHIFT
PULSE BUFFER
DFP
CURRENT SIG. AMPS
PHASE DELAYS
3 PHASE GEN
SHIFT COORD
VOLTAGE REGULATOR
MODULATOR 2
MULTIPLEXER
SYNCHRONIZER (CONT'D FROM SHEET 1)
POWER INTERFACE
RESISTANCE ISOLATOR
± 20VPS
+5VPS
COMMUTATION A2
COMMUTATION B2
COMMUTATION C2
TO F COM BUS
TO TB6.GND
TEST A
TEST B
TEST C
SEE NOTE 6
SEE NOTE 3
TO ACOM ON 5.17, 6.17, 7.17, 8.17 COMMUTATION CARDS
REMOVE INDICATED CONNECTIONS FOR G02
REVISIONS
speed variator
ERIE, PA.
36D865532AA
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

WIRE TABLE

BUS BAR CONNECTIONS

TAB	CARD CONNECTED	CIRCUIT
02	02, 04, 05, 06, 07, 08, 09.10,11,12,13, 15	-20V
02	10, 19, 20, 21, 23, 26, 28,30	-20V
04	02, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13,15	+5V
04	18, 19, 20, 21, 23, 26, 28, 30	+5V
07	02, 04, 05, 06, 07, 08, 09, 10,	-5V
31	02, 04, 05, 06, 07,08,09,10,11,12,13,15	+20V
31	18,19,20,21,23,26,28,30	+20V
32	02,04,05,06,07,08,09,10,11,12,13,15	COM
32	16, 11, 12, 15, 16 BEND	COM
01	29, 28, 30 BUS BAR	COM
17	02,04,05,06,07,08,09	F COM

BACK PLANE WIRING

#24 WHITE

19.25-02.05-11.17-12.17
18.25-02.06-04.30-05.30-06.30-07.30-08.30-09.30-10.24
02.13-10.17
02.18-13.20X
02.19-13.22X
02.20-13.24X
02.21-13.26X
04.03-19.06
04.09-19.09
04.12-19.08
04.13-05.13-06.13-10.05-13.14
04.14-19.10
04.18-04.04■
04.19-10.27
04.21-05-21-06.21-07.21-08.21-09.21-10.09
18.20-04.22-05.22-06.22-07.22-08.22-09.22-15.10
04.23-21.28
04.24-05.24-06.24-07.24-08.24-09.24-10.22-21.21
04.26-19.21-30.27-15.21
18.16-04.28-11.21
04.29-04.31■

#24 WHITE

05.03-19.11
05.09-19.13
05.12-19.12
05.14-19.14
05.18-05.04■
05.19-10.03
05.23-21.26
19.22-05.26-15.18-30.29
18.17-05.28-11.18
05.29-05.31■
06.03-19.16
06.09-19.18
06.12-19.17
06.14-19.19
06.18-06.04■
06.19-10.08
06.23-21.24
19.23-06.26-15.12-30.28
18.18-06.28-11.23
06.29-06.31■

07.03-20.06
07.09-20.09
07.12-20.08
07.13-08,13-09,13-13,12
07.14-20.10
07.18-07.04■
07.23-20.20
20.23-07.26-13.18X
07.28-12.21
07.29-07.31■

#24 WHITE

08.03-20.11
08.09-20.13
08.12-20.12
08.14-20.14
08.18-08.04■
08.23-20.21
20.24-08.26-13.17X
08.28-12.18
08.29-08.31■

09.03-20.16
09.09-20.18
09.12-20.17
09.14-20.19
09.18-09.04■
09.23-20.22
20.25-09.26-13,13X
09.28-12.23
09.29-09.31■

#24 WHITE (CONT'D)

10.06-21.30
10.18-18.24-30.17
10.19-21.22
10.23-30.23
10.25-18.21
10.26-18.23-30.30
10.29-18.22-28.03-30.05
10.30-11.30-13.05X
21.14-11.27-13.14X-13.27-30.25
18.19-11.05-13.18-15.13
11.11-12.11
11.12-12.12
11.13-12.13-30.20
18.08-11.14-12.14-28.10
11.19-11.25
11.22-11.01
11.24-13.29-30.11
11.26-13.28-30.08
11.28-30.09
11.29-30.12

12.05-13.13
12.19-12.25
12.22-12.01
12.24-13.22
12.26-13.23
12.27-13.25
12.28-13.24
12.29-13.21
12.30-13.26

13.05-15.24
13.06-15.19
13.06X-30.14-18.13
13.08-15.29
13.09-15.26
13.10-15.30
13.11-15.06
13.15X-30.16-18.14
13.16-15.09-15.23
13.17-15.28-18.12
13.19-15.17
13.20-15.25
13.29X-18.10
15.05-15.08-28.11-19.26
18.09-28.29-30.06-15.11

18.06-28.22-30.10
18.11-28.27
18.26-30.13
18.28-19.28-20.28
18.30-19.30-20.30-13.27X
19.20-30.26
19.24-28.14-30.22
21.19-28.26
28.13-30.21

+5V

26.04-26.16
02.04-18.04
04.04-19.04
05.04-20.04
06.04-21.04

COMMON

26.01-26.27
23.15-11.32
23.16-12.32
23.17-13.32
23.18-28.01

F COM

02.17-18.01-19.01-20.01-21.01

-5V

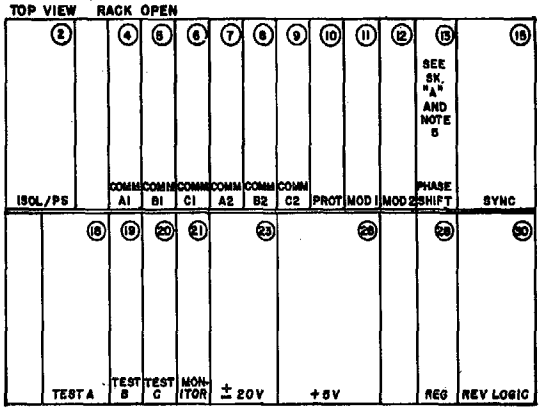
02.07-18.07-19.07-20.07-21.07

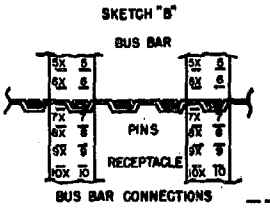
-20V

02.02-18.02
04.02-19.02
05.02-20.02

+20V

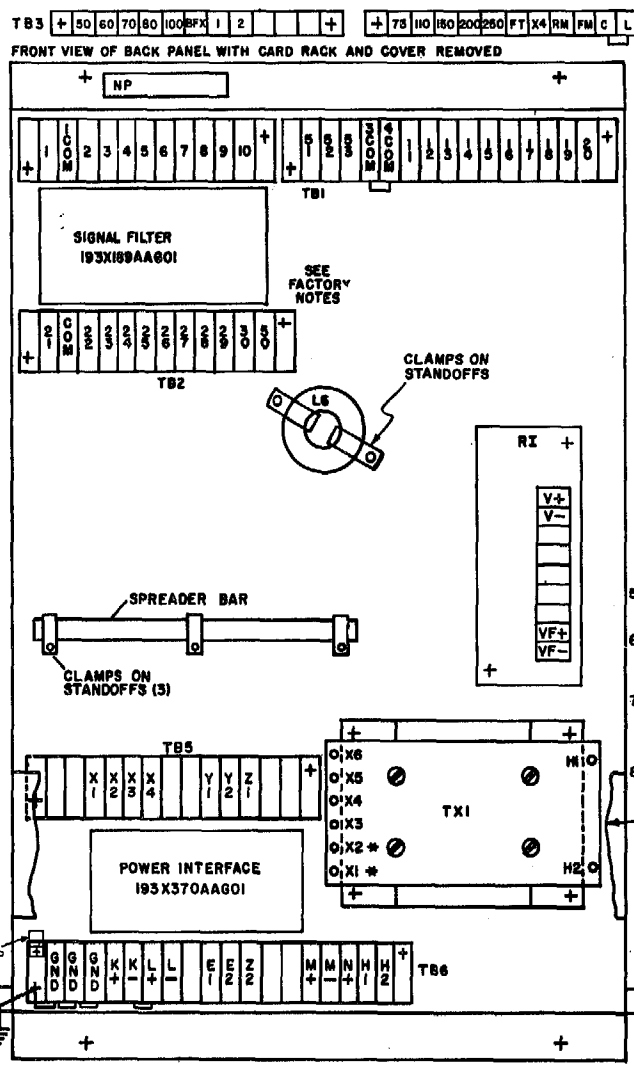
02.31-18.31
04.31-19.31
05.31-20.31





FACTORY NOTES:

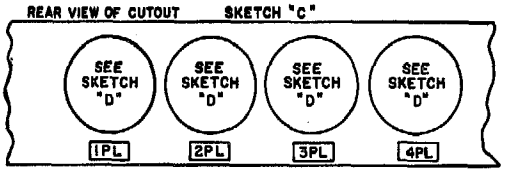
1. ASSEMBLE BUS BAR BEFORE WIRING. ALWAYS ASSEMBLE WITH THE BODY OF THE BUS TOWARD THE LOWER NUMBERED SIDE OF THE RECEPTACLE.
2. CONNECT WIRES TO TB1, TB2 & TB3 AT LOWER TERMINALS; ALL WIRES TO TB5 & TB6 AT UPPER TERMINALS.
3. *-CRIMP ALL #24 WIRES ON ONE SPADE TERMINAL.
4. TB3 HAS BOTH SCREW AND TERMIPOINT CONNECTORS.

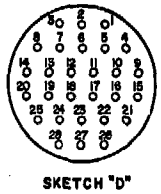


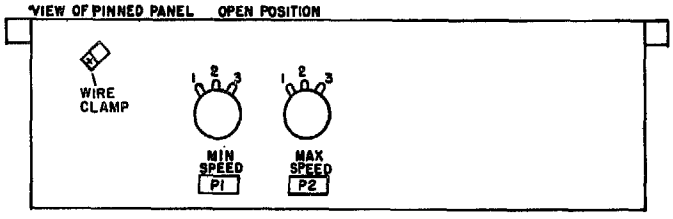


SKETCH "A"
FULL SIZE VIEW OF CARD RECEPTACLE

5. LOCATION 13 IS A 64 PIN RECEPTACLE. ALL OTHERS ARE 32 PIN.
6. DO NOT BUNDLE WIRES. DISTRIBUTE WIRES ALONG LENGTH OF SPREADER BAR.
7. CUT ALL WIRES APPROXIMATELY ONE INCH LONGER THAN THE SHORTEST POSSIBLE LENGTH.
8. ■ REMOVE FOR G02 ONLY.







UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES: 718A905
SURFACES: ✓
TOLERANCES ON MACHINED DECIMALS: ± 0.005
TOLERANCES ON MACHINED FRACTIONS: ± 1/64
ANGLES: ± 1°

FIRST MADE FOR STANDARD LINE

CONNECTION DIAGRAM
AF3122 DRIVER
331X351AAG01, G02
G01—FOR PHASE SHIFTING AF3061 INVERTERS
G02—FOR PHASE SHIFTING AF3060 INVERTERS

DRAWING NO. 36C764841AA SHT NO. 1

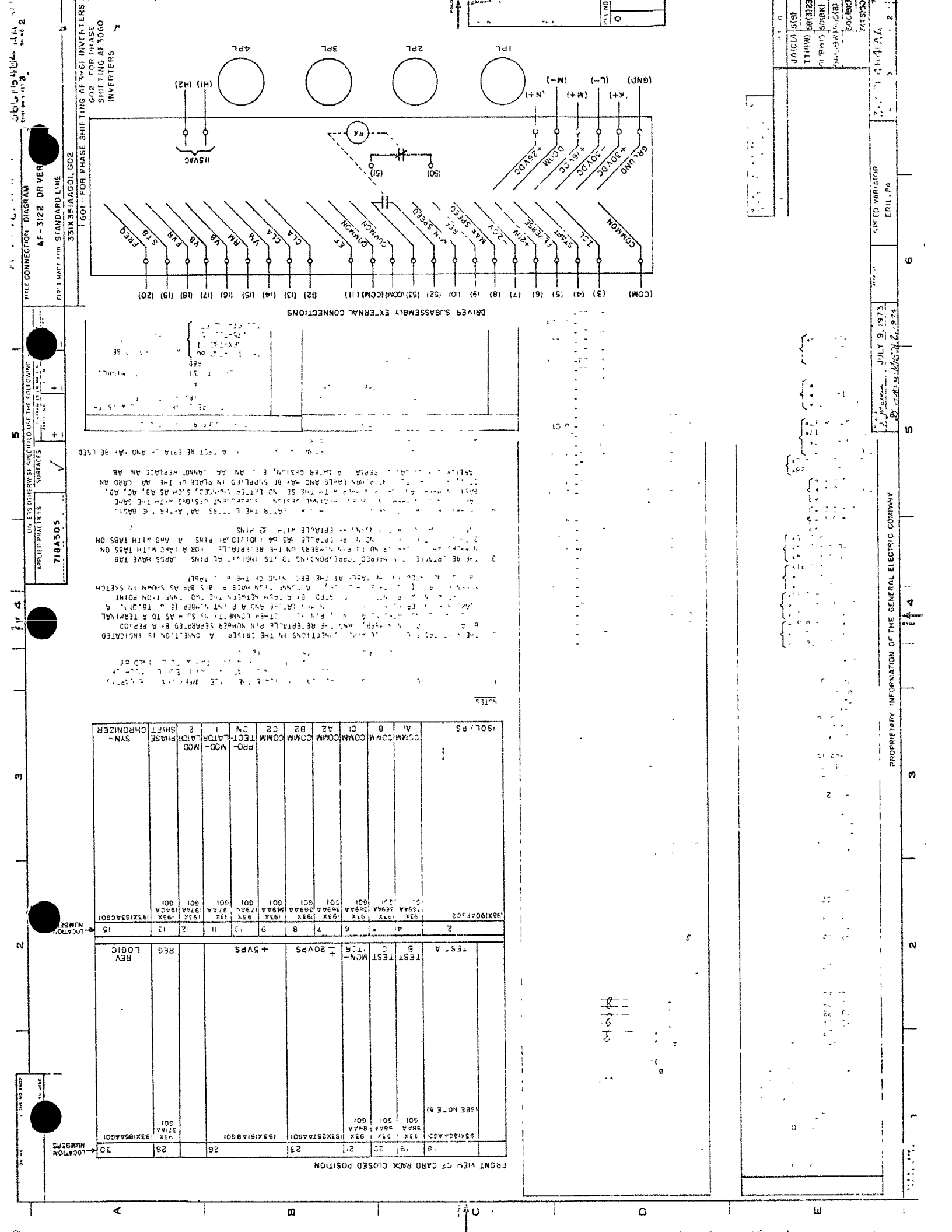
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN [illegible] JULY 9, 1973
ENGINEER [illegible]
ERIE, PA. SPEED VARIATOR
DEPT. LOCATION
36C764841AA SHT NO. 2

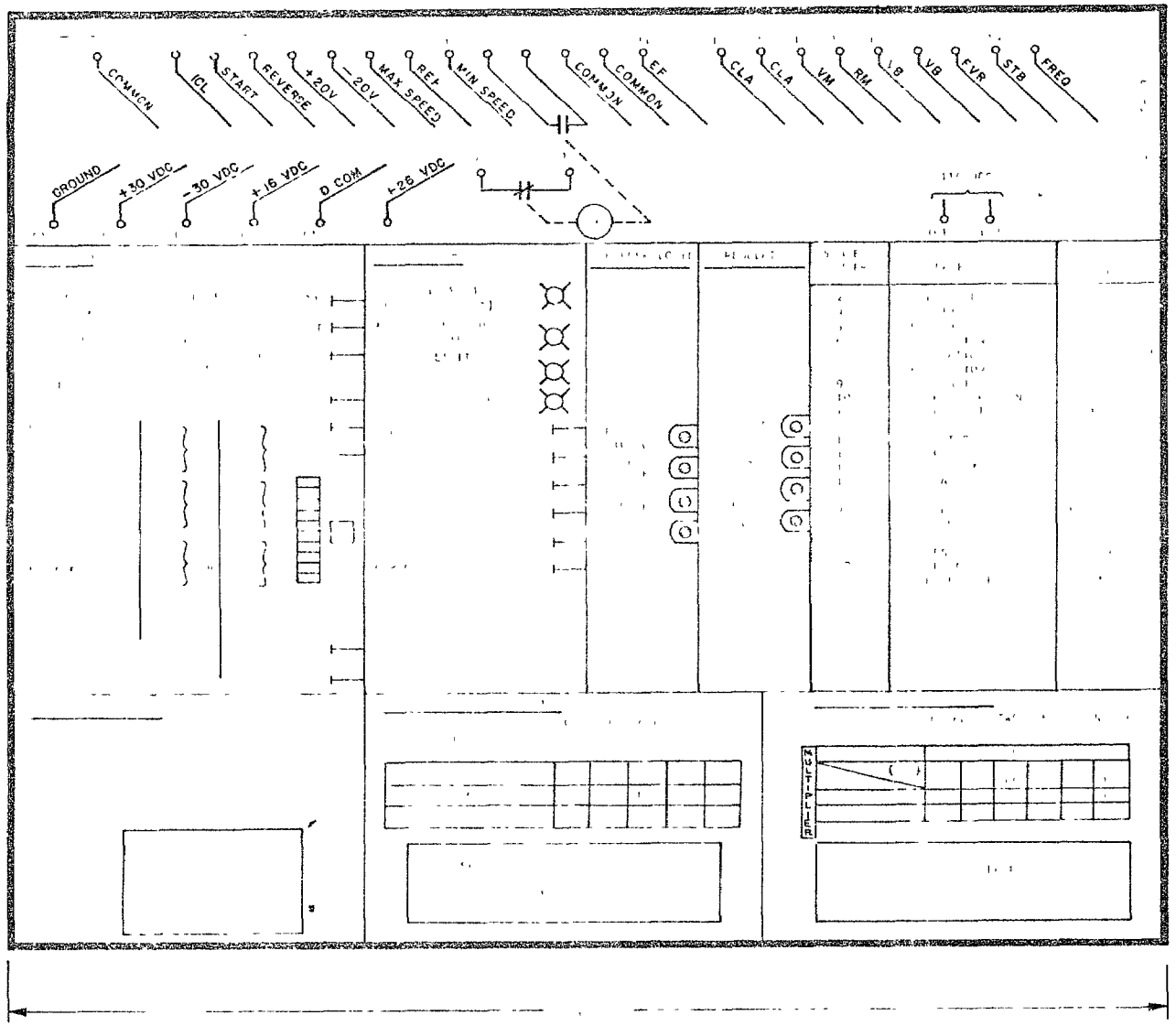
REVISIONS

No revisions are to be made to this drawing without Engineering Approval of the Speed Variator Department.

REV NO.									


TITLE CONNECTION DIAGRAM
AF-3122 DR VER
331X351AAG01, G02
1 G01 - FOR PHASE SHIFTING AF 3060 INVERTERS
G02 FOR PHASE SHIFTING AF 3060 INVERTERS
APPLIED PRACTICES
718A505
SUBJECT
SPEED VARIATOR
ERIE, PA
JULY 9, 1973
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
DRIVER SUBASSEMBLY EXTERNAL CONNECTIONS
FRONT VIEW OF CARD RACK CLOSED POSITION
LOCATION NUMBERS
COMMON
+20VDC
+30VDC
-30VDC
+15VDC
DCOM
GROUND
(COM)
(3)
(4)
(5)
(6)
(7)
(8)
(9)
(10)
(52)
(53)
(COM)(11)
(12)
(13)
(14)
(15)
(16)
(17)
(18)
(19)
(20)
FREQ
STB
FVR
VB
RM
VM
CLA
EF
115VAC
(HI) (M2)
1PL
2PL
3PL
4PL

LABEL ARTWORK &
AF 3122
STANDARD LINE
331X351AAG01, G02

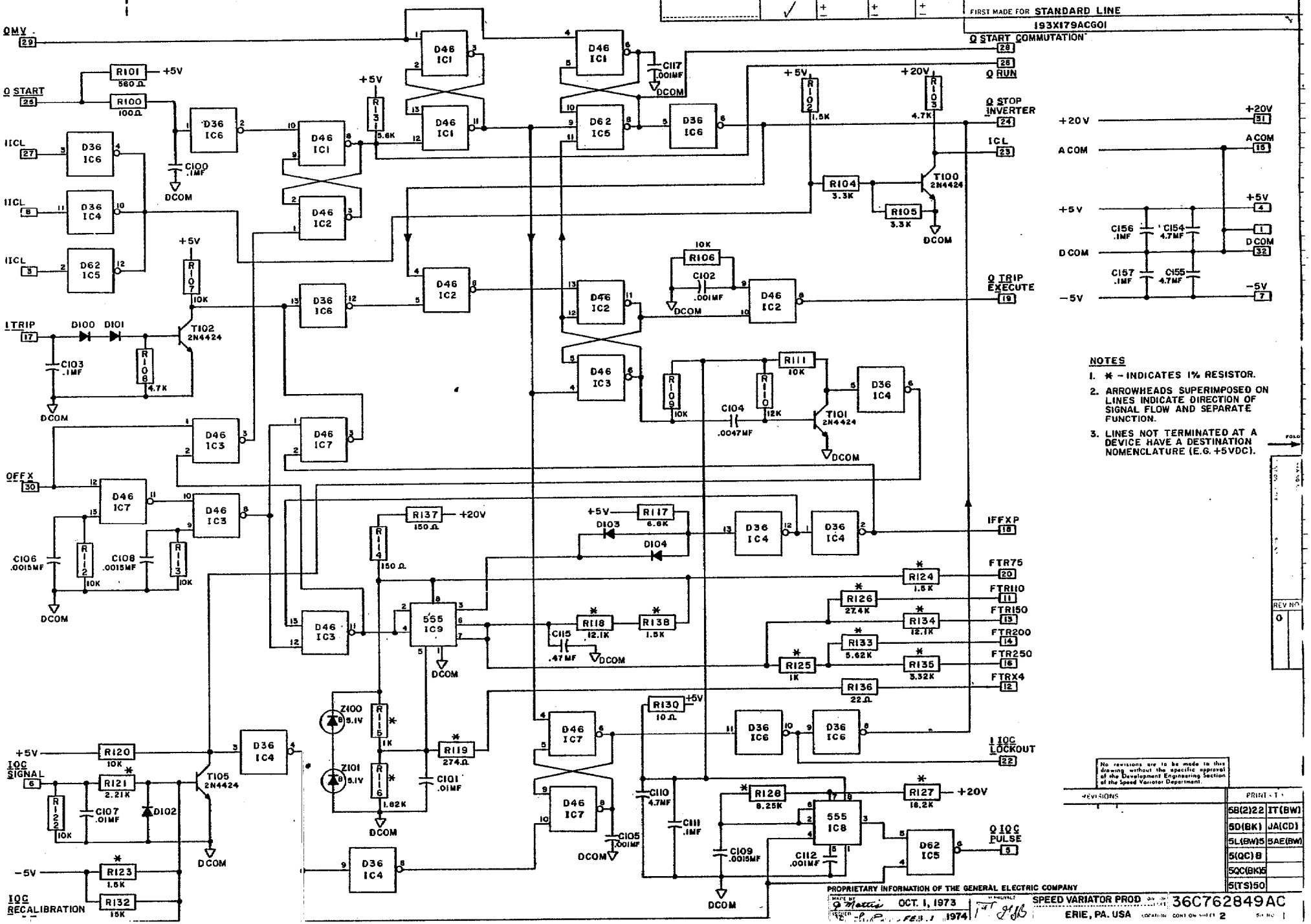


1 SIZE AS INDICATED

2 PRINT ON [illegible]

3 PLOT TO SIZE

mHuntsman June 11, 1974		SPEED VARIATOR ERIE, PA	360764841AA FL 3


GENERAL ELECTRIC
36C762849AC
TITLE ELEMENTARY DIAGRAM
DRIVE PROTECTION
193X179ACG01
Q START COMMUTATION
Q RUN
Q STOP INVERTER
ICL
Q TRIP EXECUTE
IFFXP
FTR75
FTR110
FTR150
FTR200
FTR250
FTRX4
I IOC LOCKOUT
Q IOC PULSE
QMV
Q START
IICL
ITRIP
OFFX
IOC SIGNAL
IOC RECALIBRATION
+20V
ACOM
+5V
DCOM
-5V
NOTES
1. * - INDICATES 1% RESISTOR.
2. ARROWHEADS SUPERIMPOSED ON LINES INDICATE DIRECTION OF SIGNAL FLOW AND SEPARATE FUNCTION.
3. LINES NOT TERMINATED AT A DEVICE HAVE A DESTINATION NOMENCLATURE (E.G. +5VDC).
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR PROD
ERIE, PA. USA
36C762849AC

GENERAL ELECTRIC

36C762849AC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING: APPLIED PRACTICES 71GASIG; SURFACES ✓; TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ± DECIMALS ± ANGULAR ±

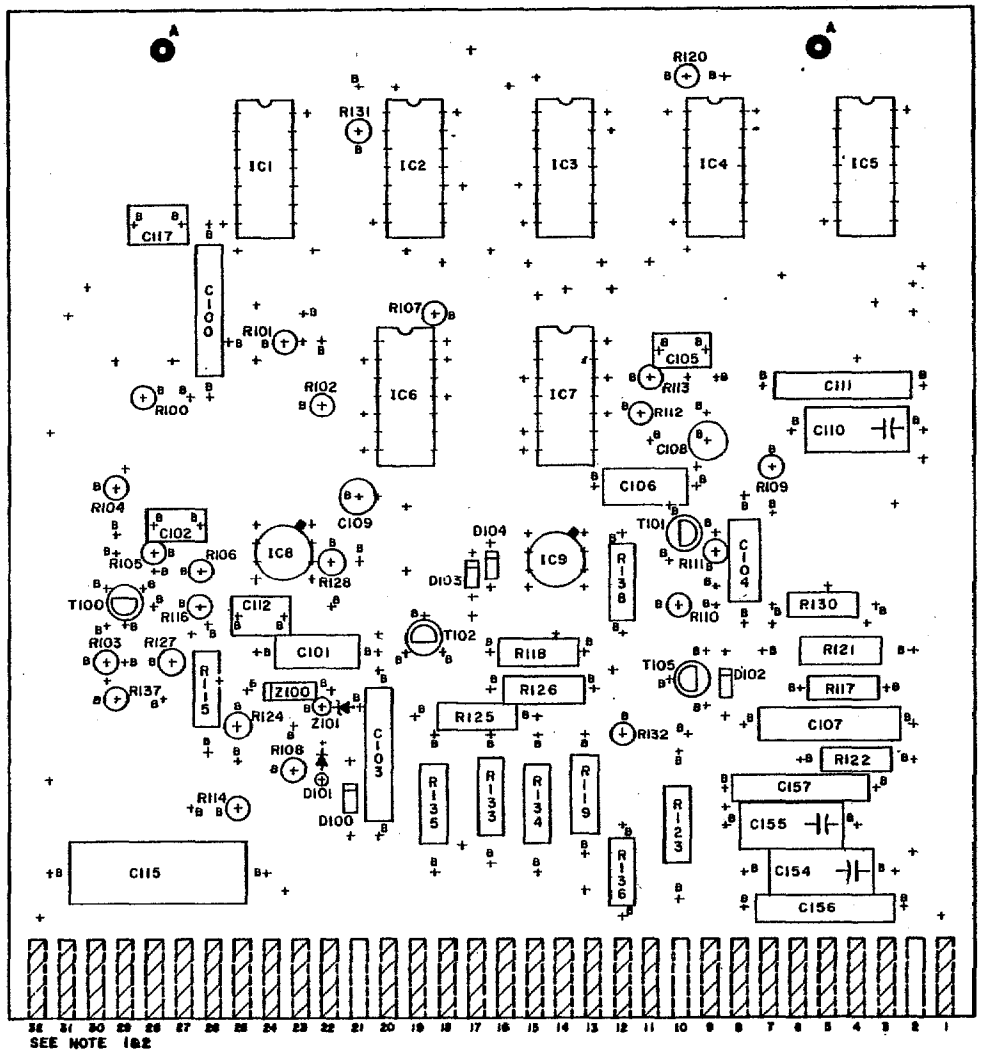
TITLE PRINTED CIRCUIT DIAGRAM
DRIVE PROTECTION
FIRST MADE FOR STANDARD LINE
193X179ACG01

CONT ON SHEET FL. SH NO 2

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.040	130

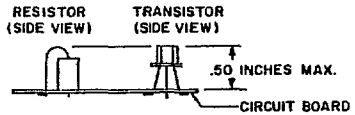


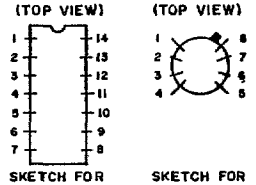
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $6.500^{+.000}_{-.018}$ X $5.150^{+.002}_{-.008}$
4. ALL TRANSISTORS AND RESISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

TRANSISTOR LEAD SKETCHES







2X SCALE	36A352983CA005 REAR ETCHED BOARD DWG.	36A352982CA005 FRONT ETCHED BOARD DWG.	1 FRONT SPACE UNIT REQUIREMENT	0 BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varrator Department

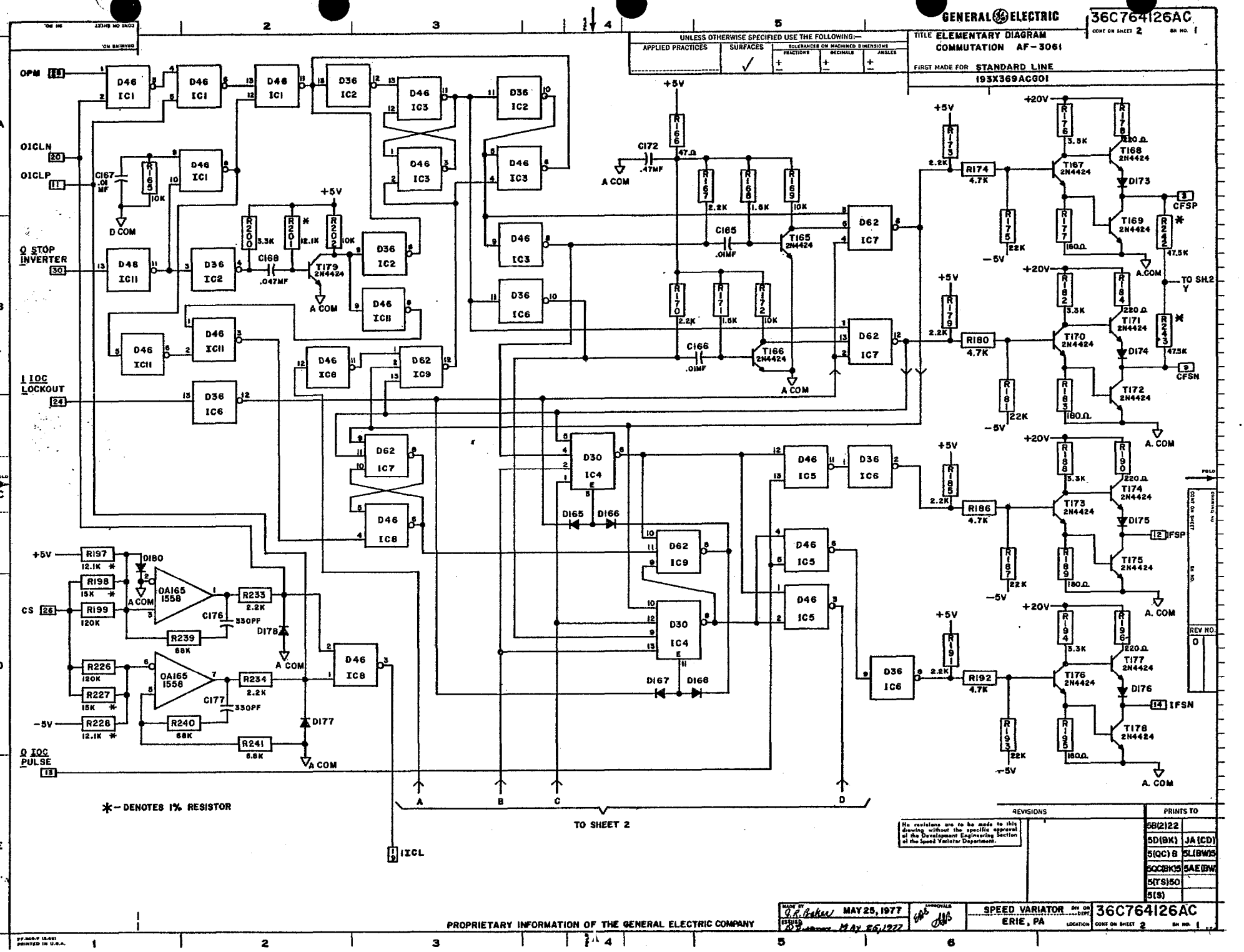
REV NO. 0

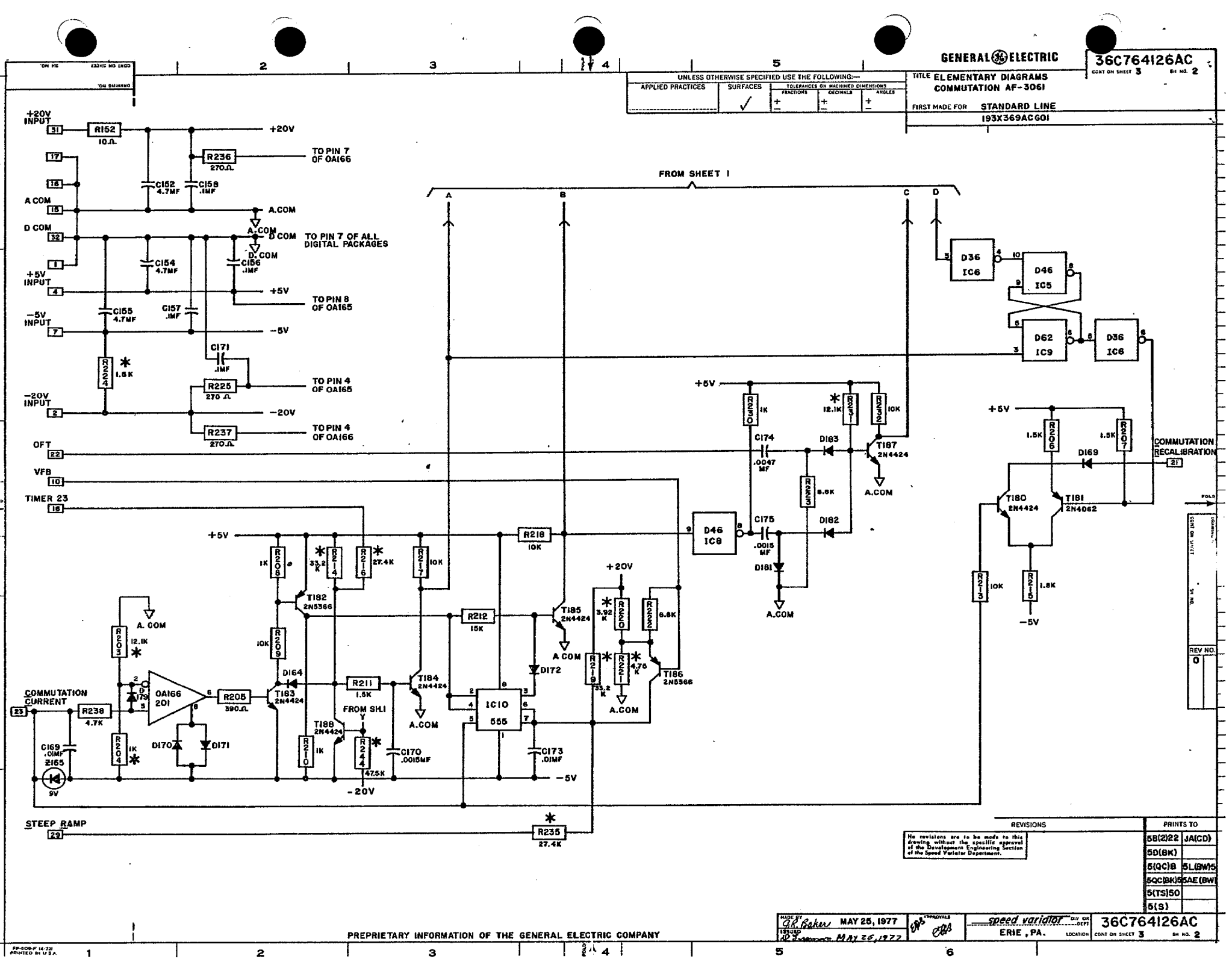
PRINTS TO

50C(B)	5 (B)
58(2)22	8D(BK)
5L(BW)5	~~8D(BK)~~
500(BK)5	5AE(BW)
5(TS)50	JA(CD)
1T(BW)	

DRAWN BY G. Mazzei OCT. 3, 1973
DESIGNED E. Lukow FEB 1, 1974
APPROVED J. Hall
SPEED VARIATOR PROD DIV OR DEPT ERIE, PA. USA LOCATION
36C762849AC CONT ON SHEET FL SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764126AC
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
COMMUTATION AF-3061
FIRST MADE FOR STANDARD LINE
193X369ACG01
OPM 15
OICLN 20
OICLP 11
Q STOP INVERTER 30
I IOC LOCKOUT 24
CS 26
Q IOC PULSE 13
D COM
A COM
-5V
+5V
+20V
CFSP 8
CFSN 9
TO SH.2
IFSP 12
IFSN 14
IICL 19
*— DENOTES 1% RESISTOR
TO SHEET 2
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
58(2)22
5D(BK) JA(CD)
5(QC) B 5L(BW)5
5QC(BK)5 5AE(BW)
5(TS)50
5(S)
MADE BY J.R. Baker MAY 25, 1977
ISSUED MAY 26,1977
SPEED VARIATOR
ERIE, PA
36C764126AC
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764126AC
CONT ON SHEET 3
SH NO. 2
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAMS
COMMUTATION AF-306I
FIRST MADE FOR STANDARD LINE
193X369AC601
+20V INPUT
R152 10Ω
+20V
R236 270Ω
TO PIN 7 OF OA166
C152 4.7MF
C158 .1MF
A COM
A.COM
D COM
TO PIN 7 OF ALL DIGITAL PACKAGES
C154 4.7MF
C156 .1MF
+5V INPUT
+5V
TO PIN 8 OF OA165
C155 4.7MF
C157 .1MF
-5V INPUT
-5V
C171 .1MF
R224 1.5K
R225 270Ω
TO PIN 4 OF OA165
-20V INPUT
-20V
R237 270Ω
TO PIN 4 OF OA166
OFT
VFB
TIMER 23
FROM SH 1
D36 IC6
D46 IC5
D62 IC9
D36 IC6
+5V
R200 1K
R201 12.1K
R202 10K
C174 .0047 MF
D183
T187 2N4424
R203 6.8K
D182
C175 .0015 MF
D46 IC8
D181
A.COM
COMMUTATION RECALIBRATION
R206 1.5K
R207 1.5K
D169
T180 2N4424
T181 2N4062
R213 10K
R214 1.8K
-5V
R218 10K
R208 1K
R214 33.2K
R216 27.4K
R217 10K
T182 2N5366
R212 15K
T185 2N4424
+20V
R209 10K
R215 3.92K
R220 6.8K
R219 33.2K
R221 4.75K
T186 2N5366
D172
D164
R211 1.5K
T184 2N4424
T183 2N4424
R203 12.1K
A. COM
COMMUTATION CURRENT
R238 4.7K
OA166 201
R205 390Ω
D179
D170
D171
R204 1K
C169 .01MF
Z165 9V
T188 2N4424
R210 1K
R244 47.5K
-20V
C170 .0015MF
IC10 555
C173 .01MF
STEEP RAMP
R235 27.4K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
58(2)22 JA(CD)
5D(8K)
5(QC)8 5L(BW)5
5QC(BK)6 5AE(BW)
5(TS)50
5(S)
MADE BY A.R. Baker MAY 25, 1977
ISSUED
APPROVALS
speed variator
ERIE, PA.
36C764126AC
CONT ON SHEET 3
SH NO. 2
PREPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
FP-809-F 14-72
PRINTED IN U.S.A.

GENERAL ELECTRIC

36C764126AC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES: 718A815 | SURFACES: ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: PRINTED CIRCUIT DIAGRAM
COMMUTATION AF-3061
FIRST MADE FOR: STANDARD LINE
193X369ACG01

CONT ON SHEET: FL. SH NO: 3

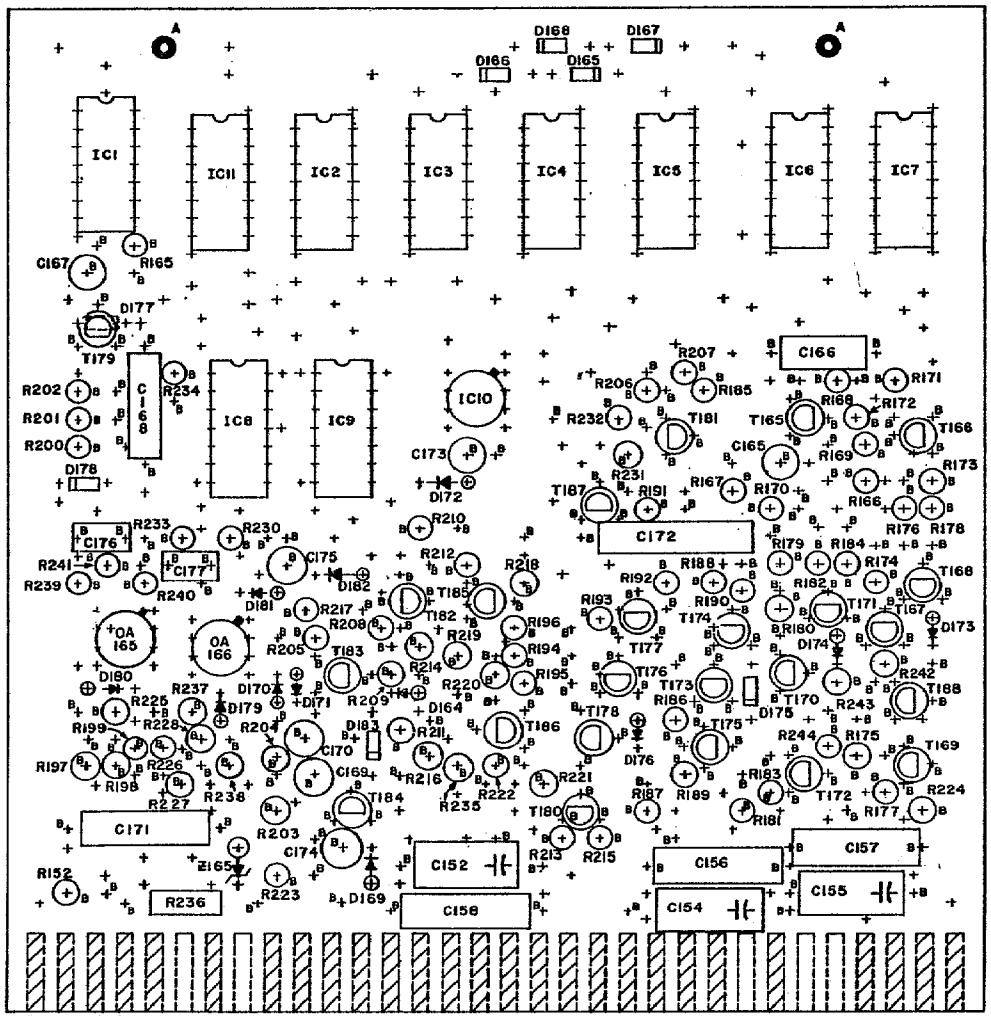
HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

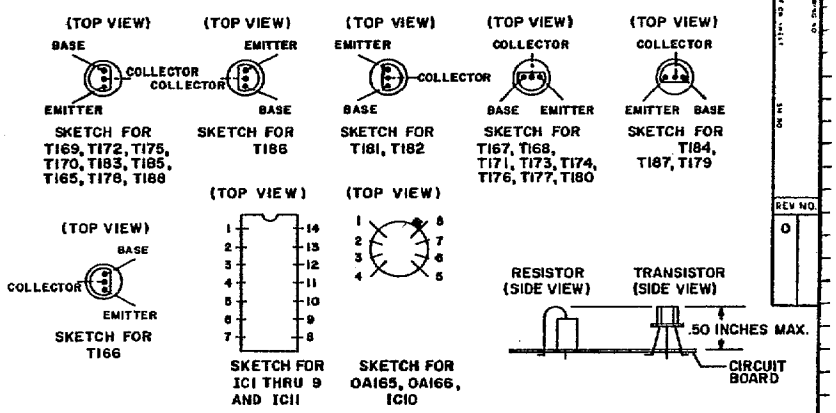
LOC.	DIA.	QUAN.
A	.157	2
B	.040	269

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.016}$ X $5.130^{+.002}_{-.008}$



TRANSISTOR LEAD SKETCHES



REV NO. 0

2X	36A353788AB005	36A353789AB005	1	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	SPACE UNIT	REQUIREMENT

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

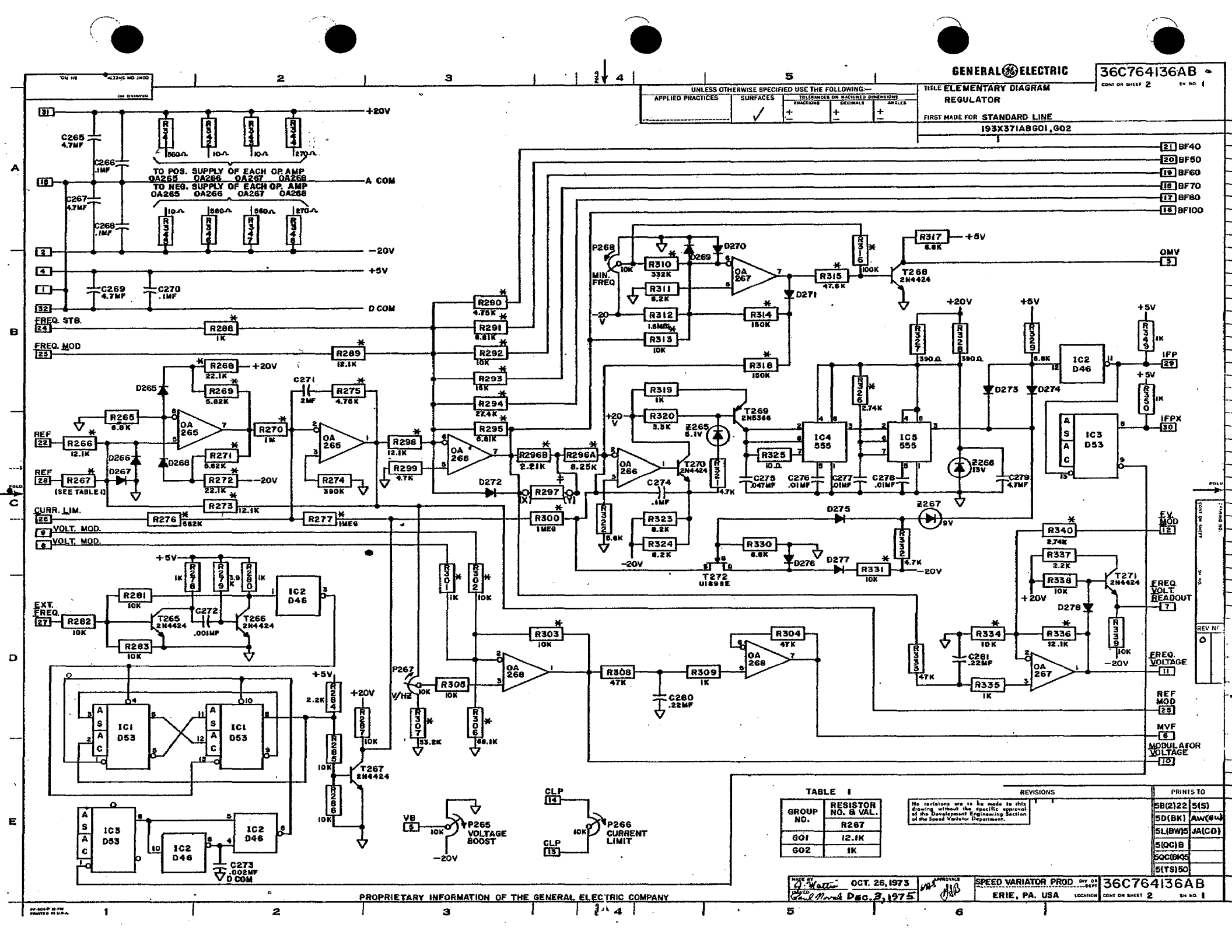
PRINTS TO

58(2)22	
5D(BK)	JA(CD)
5(QC)B	5L(BW)5
5QC(BK)5	5AE(BW)
5(TS)50	
5(S)	

DRAWN: J.R. Baker MAY 25, 1977
CHECKED: [illegible] MAY 25, 1977
APPROVALS: [illegible]
SPEED VARIATOR PROD DIV. OR DEPT
ERIE, PA. LOCATION

36C764126AC
CONT ON SHEET: FL. SH NO: 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764136AB
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
REGULATOR
FIRST MADE FOR STANDARD LINE
193X371ABG01, G02
+20V
A COM
−20V
+5V
D COM
FREQ. STB.
FREQ. MOD
REF
CURR. LIM.
VOLT. MOD.
EXT. FREQ.
TO POS. SUPPLY OF EACH OP AMP
OA265 OA266 OA267 OA268
TO NEG. SUPPLY OF EACH OP. AMP
OA265 OA266 OA267 OA268
BF40
BF50
BF60
BF70
BF80
BF100
OMV
IFP
IFPX
EV MOD
FREQ. VOLT READOUT
FREQ. VOLTAGE
REF MOD
MVF
MODULATOR VOLTAGE
P265 VOLTAGE BOOST
P266 CURRENT LIMIT
P267 V/Hz
P268 MIN. FREQ
TABLE I
GROUP NO.
RESISTOR NO. & VAL.
R267
G01 12.1K
G02 1K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
PRINTS TO
5B(2)22 5(S)
5D(BK) AW(BL)
5L(BW)5 JA(CD)
5(QC)B
5QC(BK)5
5(TS)50
OCT. 26, 1973
DRAWN Dec. 3, 1975
SPEED VARIATOR PROD
ERIE, PA. USA
36C764136AB
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

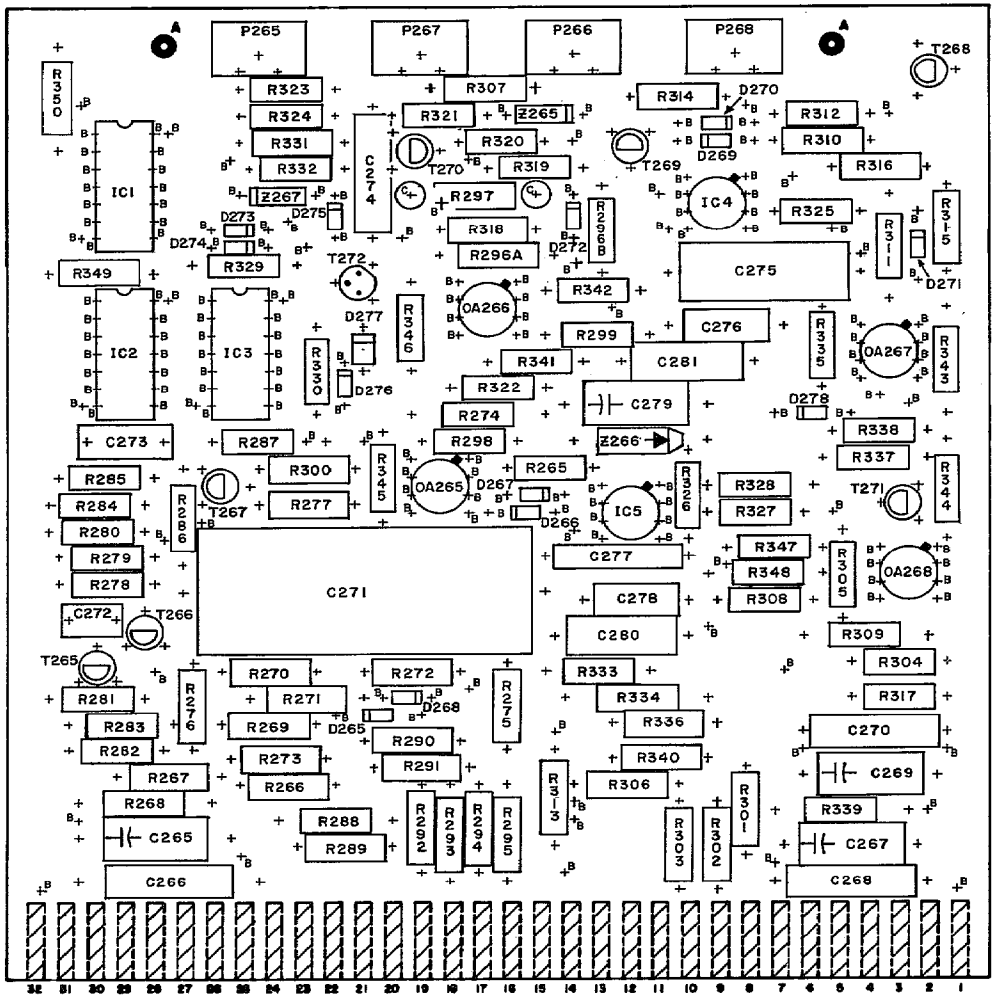
GENERAL ELECTRIC

36C764136AB
CONT ON SHEET FL. SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
710A816	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
REGULATOR CARD
FIRST MADE FOR STANDARD LINE AF-3061
193X371ABG01,G02



HOLE TABULATION

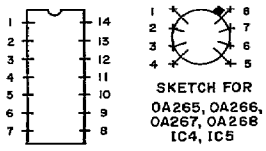
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.032	168
C	.081	2

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND OP AMPS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.
5. FOR-G02 ONLY
   STAMP ADHESIVE LABEL-G02. ADD LABEL AS SUFFIX TO 193X371AA IN PLACE OF G01 STAMPED ON PRINTED CIRCUIT BOARD.

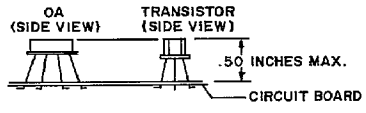
IC & OA LEAD SKETCHES



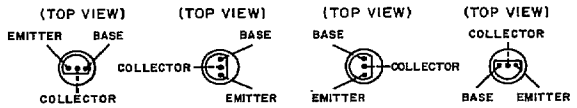
SKETCH FOR IC1, IC2, IC3

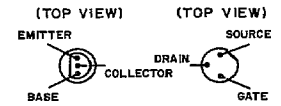
SKETCH FOR OA265, OA266, OA267, OA268 IC4, IC5

COMPONENT HEIGHT SKETCH



TRANSISTOR LEAD SKETCHES





2X	36A353697AC005	36A353698AC005	1	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

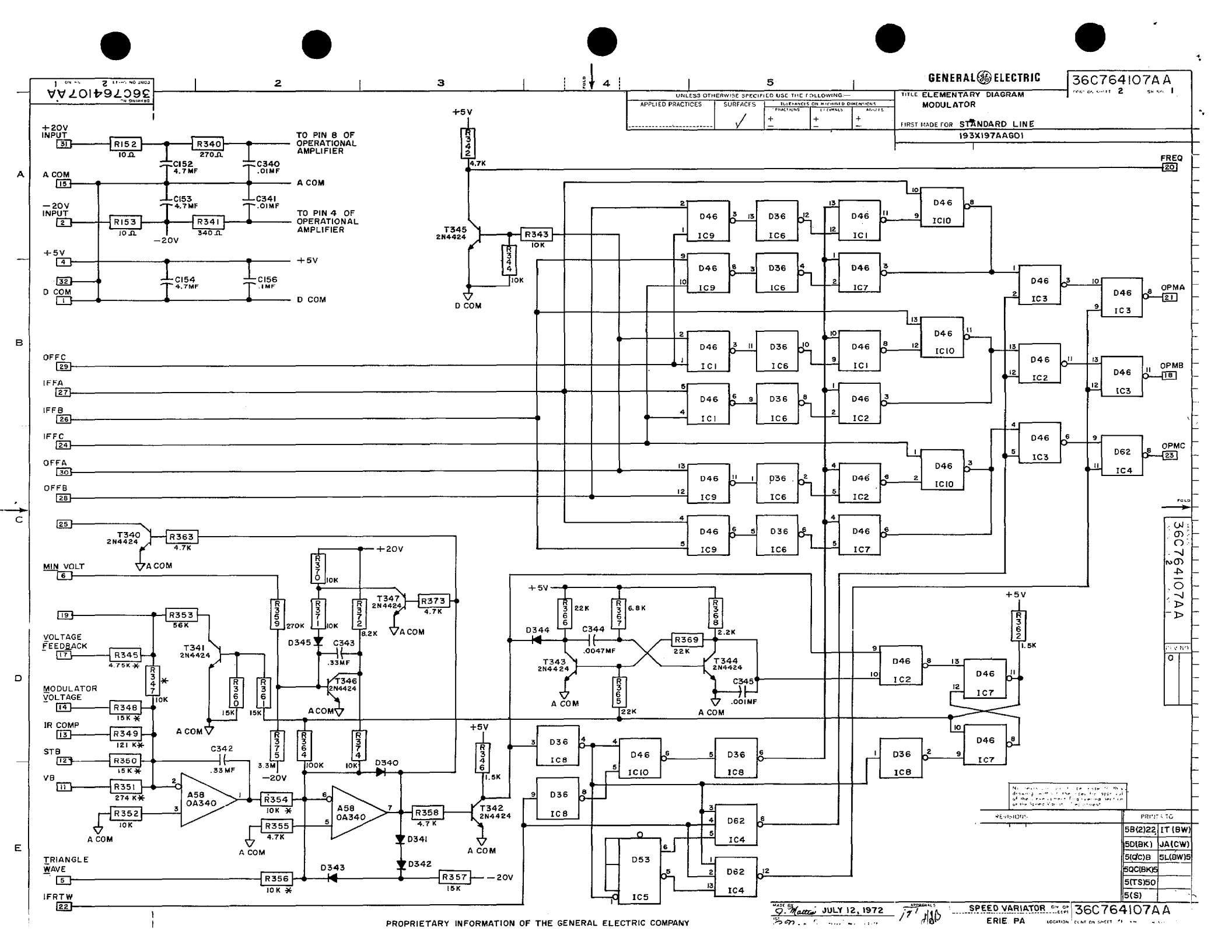
1	M. Cotty Nov. 30, 1978	PER SV-883#37	

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	TS(50)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

DRAWN BY A. Martin NOV. 21, 1973
ISSUED BY M. Cotty Nov. 30, 1978
APPROVALS ERS JHB
SPEED VARIATOR PROD. DIV. OR DEPT. ERIE, PA. USA LOCATION
36C764136AB
CONT ON SHEET FL. SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C764107AA
TITLE ELEMENTARY DIAGRAM MODULATOR
FIRST MADE FOR STANDARD LINE
193X197AAG01
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
+20V INPUT
-20V INPUT
A COM
D COM
+5V
TO PIN 8 OF OPERATIONAL AMPLIFIER
TO PIN 4 OF OPERATIONAL AMPLIFIER
R152 10Ω
R153 10Ω
R340 270Ω
R341 340Ω
C152 4.7MF
C153 4.7MF
C154 4.7MF
C156 1MF
C340 .01MF
C341 .01MF
FREQ
OFFC
IFFA
IFFB
IFFC
OFFA
OFFB
MIN VOLT
VOLTAGE FEEDBACK
MODULATOR VOLTAGE
IR COMP
STB
VB
TRIANGLE WAVE
IFRTW
OPMA
OPMB
OPMC
T340 2N4424
T341 2N4424
T342 2N4424
T343 2N4424
T344 2N4424
T345 2N4424
T346 2N4424
T347 2N4424
A58 0A340
D46 IC9
D46 IC10
D46 IC1
D46 IC2
D46 IC3
D46 IC7
D36 IC6
D36 IC8
D62 IC4
D62 IC3
D62 IC4
D53 IC5
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR
ERIE PA
JULY 12, 1972
36C764107AA
REVISIONS
PRINTS TO
5B(2)22 IT (BW)
5D(BK) JA(CW)
5(QC)8 5L(BW)5
5QC(BK)5
5(TS)50
5(S)

GENERAL ELECTRIC

36C764107AA CONT ON SHEET FL SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

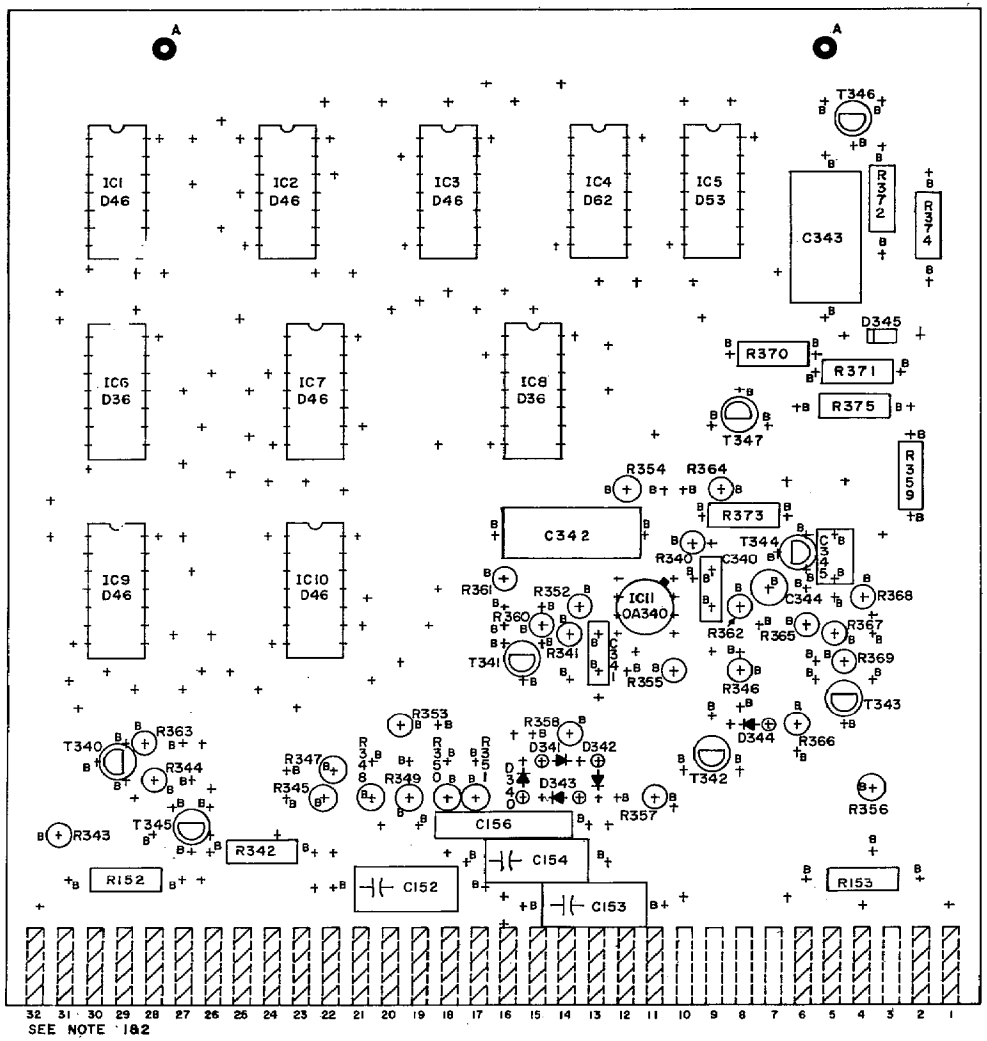
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
MODULATOR
FIRST MADE FOR STANDARD LINE
193X197AAG01

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

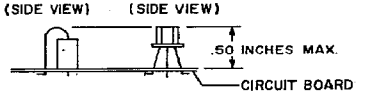
LOC.	DIA.	QUAN.
A	.157	2
B	.040	120



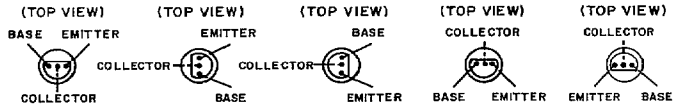
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS AND RESISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

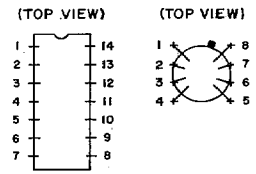
RESISTOR (SIDE VIEW) TRANSISTOR (SIDE VIEW)



TRANSISTOR LEAD SKETCHES



IC LEAD SKETCHES



2X	36A352956AA005	36A352955AA005	1	—
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE	BACK UNIT REQUIREMENT

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

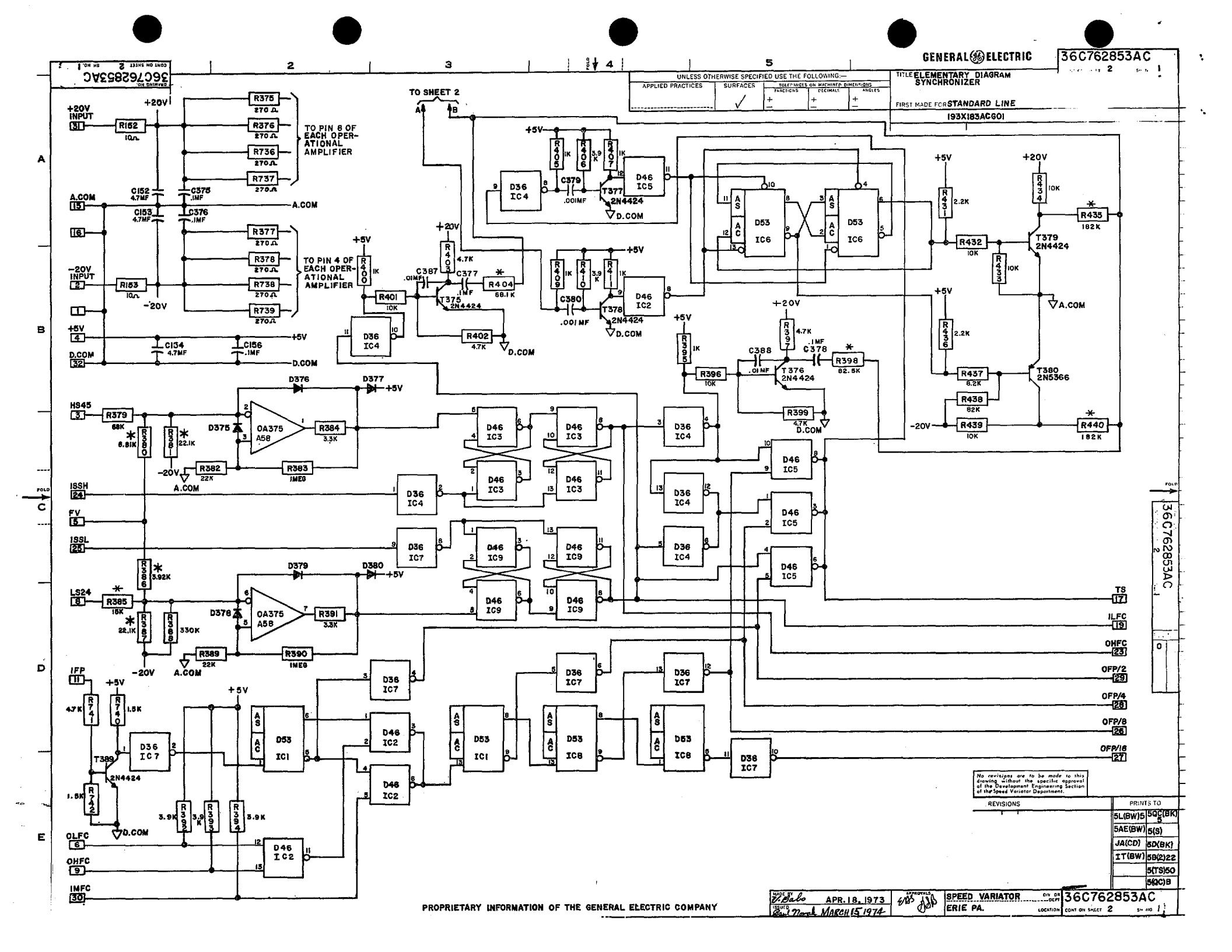
PRINTS TO	
5B(2)22	IT(BW)
5D(BK)	JA(CD)
5(QC)8	5L(BW)5
5QQ(BK)5	
5(TS)50	
5(S)	

MADE BY D. Mattis AUG. 25, 1972 APPROVAL

SPEED VARIATOR ERIE, PA.

36C764107AA CONT ON SHEET FL SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36C762853AC
TITLE ELEMENTARY DIAGRAM SYNCHRONIZER
FIRST MADE FOR STANDARD LINE
193X183ACG01
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TO SHEET 2
+20V INPUT
-20V INPUT
A.COM
D.COM
+5V
-20V
TO PIN 8 OF EACH OPERATIONAL AMPLIFIER
TO PIN 4 OF EACH OPERATIONAL AMPLIFIER
HS45
LS24
ISSH
ISSL
FV
IFP
OLFC
OHFC
IMFC
TS
ILFC
OHFC
OFP/2
OFP/4
OFP/8
OFP/16
D53 IC1
D53 IC6
D53 IC8
D46 IC1
D46 IC2
D46 IC3
D46 IC4
D46 IC5
D46 IC9
D36 IC4
D36 IC7
OA375 A58
T375 2N4424
T376 2N4424
T377 2N4424
T378 2N4424
T379 2N4424
T380 2N5366
T389 2N4424
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR
ERIE PA.
APR. 18, 1973
MARCH 15 1974
PRINTS TO
5L(BW)5
5AE(BW)
JA(CD)
IT(BW)
5QC(BK) 5
5(S)
5D(BK)
5B(2)22
5(TS)50
5(QC)8

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM
SYNCHRONIZER

FIRST MADE FOR STANDARD LINE
193X183AC901

FROM SHEET 1

D387 C386 .47MF R412 15K
R420 22.1K R419 220K R416 10K R414 10K R415 4.7K R417 10K R418 4.7K R413 100K
C381 .01MF
T388 U1898E
OA376 A58 OA377 A58 OA378 A58
A.COM D.COM
R421 10K
+20V R428 2.21K R429 15K R430 100K
C382 .001MF R427 100K
TRIANGLE WAVE 13
T383 2N4424 T384 2N4424 D382
D381 +5V T382 2N5366 R424 100K
T381 2N4424 R422 10K D386
R423 3.3K R425 33K R426 5.62K
-20V

CFA 22 R441 4.75K R443 4.7K R442 18.2K OA376 A58 CSA 21
CFB 20 R444 4.75K R446 4.7K R445 18.2K OA377 A58 CSB 18
CFC 14 R447 4.75K R449 4.7K R448 18.2K OA378 A58 CSC 12

+20V R450 22Ω
R451 2.21K R452 6.81K R453 6.81K R454 10K C383 .01MF
C385 4.7MF
T385 2N5366 T386 2N5366
R456 1K R457 1K
R455 475Ω R458 332Ω
C384 .1MF R459 3.3K
R460 2.2K D383 OFT 10 +5V
T387 2N4424 D384 D385 A.COM
R461 1.2K -20V

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

5L(BW)5	5QC(BK)
5AE(BW)	5(S)
JA(CD)	5D(BK)
IT(BW)	5B(2)22
	5(TS)50
	5(QC)B

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY V. Balo APR. 18, 1973 APPROVALS
SPEED VARIATOR ERIE PA.
DWG OR DEPT NO 36C762853AC
LOCATION CONT'D ON SHEET 3 SH NO 2

GENERAL ELECTRIC

36C762853AC

CONT ON SHEET FL SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM SYNCHRONIZER

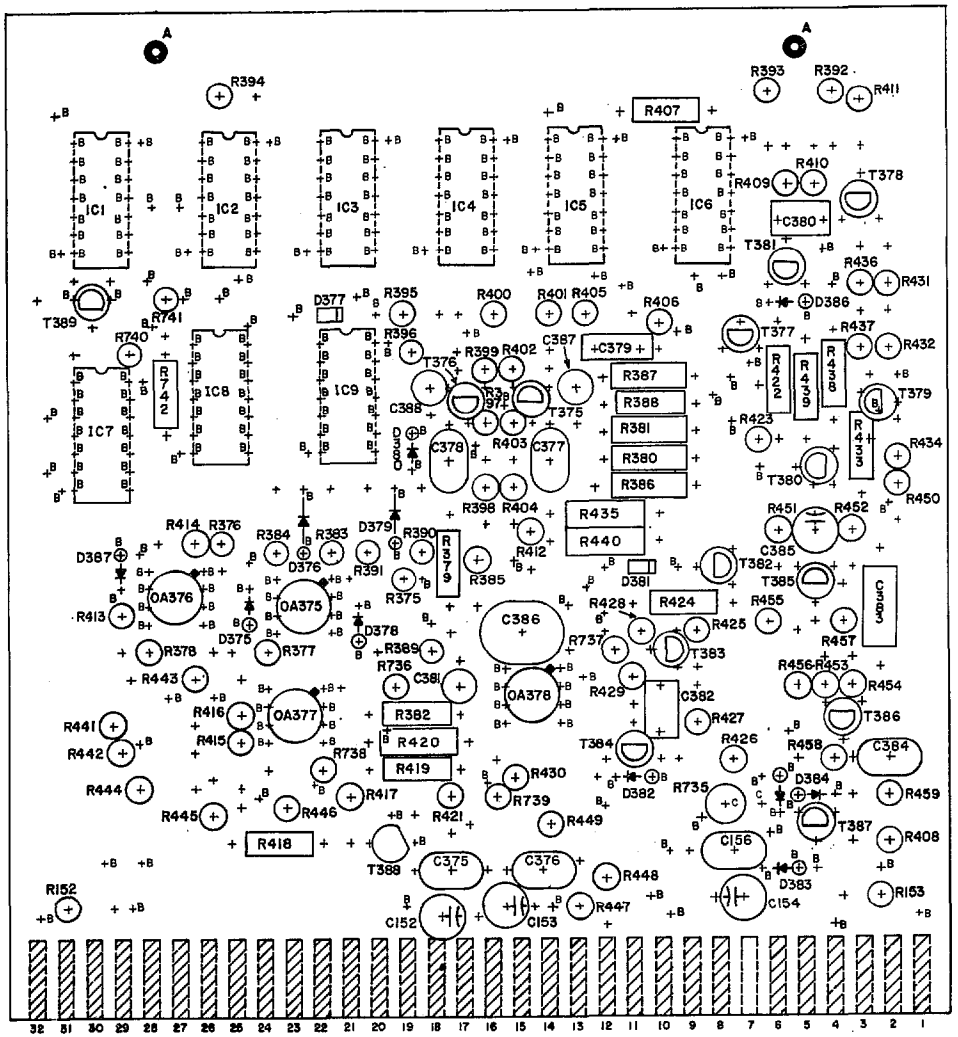
FIRST MADE FOR STANDARD LINE

193X183ACG01

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

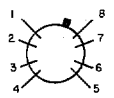
LOC.	DIA.	QUAN.
A	.157	2
B	.032	272
C	.052	2



NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

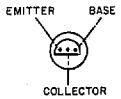
(TOP VIEW)



SKETCH FOR 0A375, 0A376, 0A377, 0A378

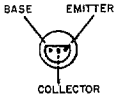
TRANSISTOR LEAD SKETCHES

(TOP VIEW)



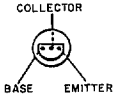
SKETCH FOR T375, T385, T387, T389

(TOP VIEW)



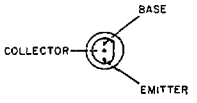
SKETCH FOR T376, T386

(TOP VIEW)



SKETCH FOR T377, T378, T381, T384

(TOP VIEW)



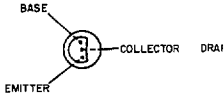
SKETCH FOR T380

(TOP VIEW)



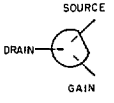
SKETCH FOR T382, T383

(TOP VIEW)



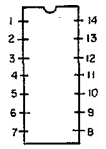
SKETCH FOR T379

(TOP VIEW)



SKETCH FOR T388

(TOP VIEW)



SKETCH FOR IC PACKAGES

2X	36A352972AC005	36A352973AC005	2	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT	BACK
			SPACE UNIT REQUIREMENT	

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV NO. 0

PRINTS TO

5QC(S)	5(S)
5AE(BW)	5D(BK)
JA(CD)	5B(2)
IT(BW)	5T8(5D
5L(BW)5	5QC(B)

MADE BY C. Mattice MAY 18, 1973

ISSUED C. Parker MARCH 15, 1974

APPROVALS EAS JHB

SPEED VARIATOR DIV OR DEPT ERIE LOCATION

36C762853AC

CONT ON SHEET FL SH NO 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C762855AB

CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
REVERSE LOGIC

FIRST MADE FOR STANDARD LINE
193X185AAG01

+20V INPUT 31
-20V INPUT 2
A. COM 18
+5V 4
D. COM 1, 32
R152 10Ω
R153 10Ω
+20V
-20V
R543 270Ω
R475 270Ω
R544 270Ω
C152 4.7MF
C153 4.7MF
C154 4.7MF
C475 .1MF
C476 .1MF
C156 .1MF
C479 .1MF
C480 .1MF
TO POS SUPPLY OF EACH OPERATIONAL AMPLIFIER
TO NEG SUPPLY OF EACH OPERATIONAL AMPLIFIER
A COM
+5V
D. COM

IFFY 14
IFFZ 16
OMV 5
IFFXP 17
Q46 IC6
D46 IC6
D36 IC17
D46 IC18
D46 IFWD IC18
D46 OFWD IC18
D46 IC12
D36 IC17
OFFC 12
IFFC 11
IFFB 8
OFFB 9
TO SHEET 2 A B

OREV 13
R478 100Ω
C477 .1MF
D. COM
D36 IC17
D46 IC11
D46 OFWD IC11
D46 IFWD IC11
IFP 6
R479 2.7K
D46 IC6

REF CLAMP 30
R480 10K
T475 2N4424
R481 15K
R482 2.7K
R485 22K
-20V
A COM
T485 2N5366
R530 12.1K *
D475
D488
D489
R532 6.8K
R490 15K
R491 8.2K
R492 8.2K
R542 15K *
D490
D491
R493 15K
-20V
D492
D493
P478 TIME DOWN 10K
P477 TIME UP 10K
R539 332K
R540 332K
R541 182K
C481 4.0MF
R494 220Ω
R495 220Ω
A58 IC16
TIMED REF 10
A.COM

NEGATIVE REF 19
R528 27K
R529 3.32K *

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

11	1-15-74 D.Morrell	CHG. PER.	SV-874-P32
12	MAR 16 1974 D.Morrell	CHG. PER.	SV-874-P32A

PRINTS TO

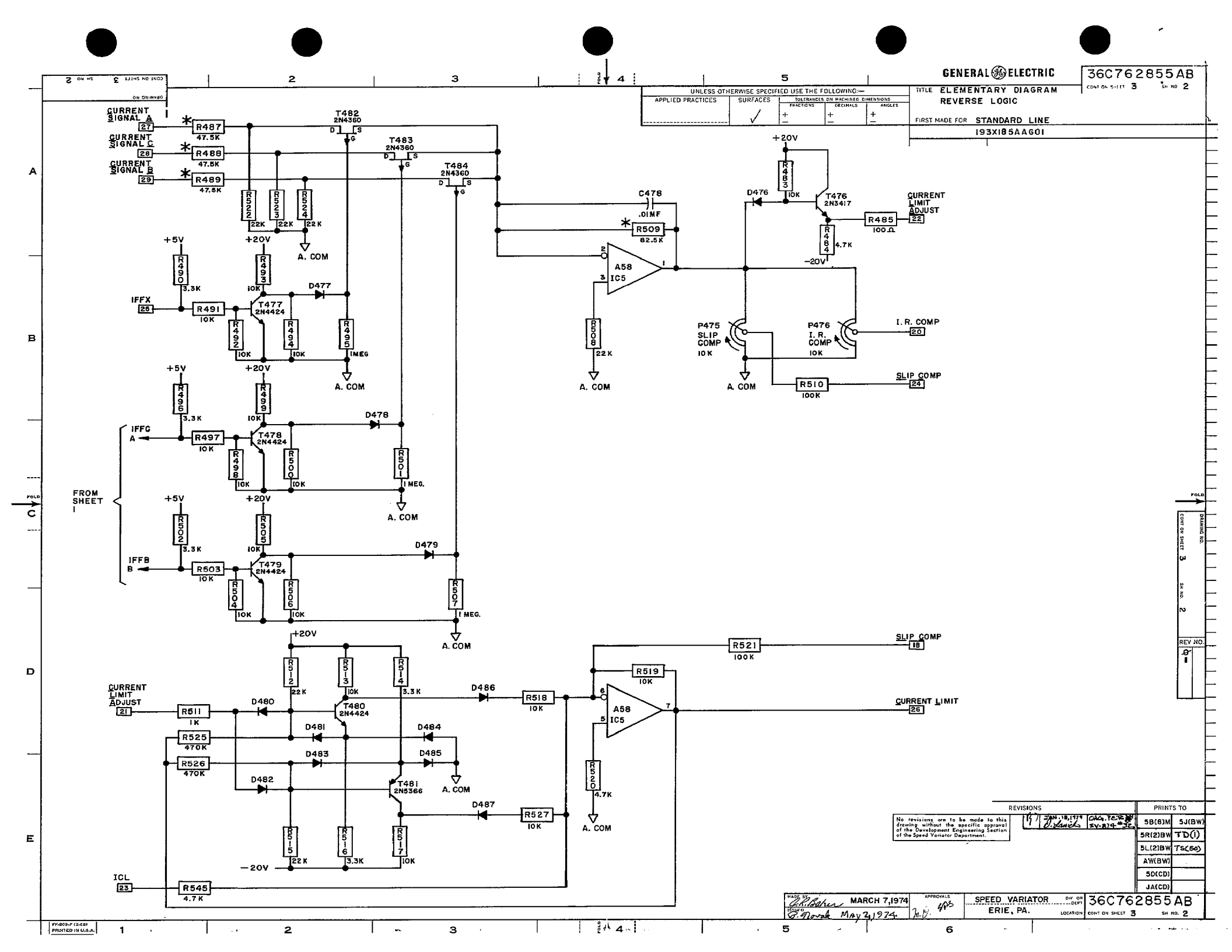
5B(8)M	5J(BW)
5R(2)BW	TD(1)
5L(2)BW	TSC(2)
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY G.P.Reiter MARCH 7,1974
DRAWN B.Morrell May 2,1974
APPROVALS M.O. [illegible]

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

36C762855AB
CONT ON SHEET 2 SH NO 1

FP-5029-F (2-68) PRINTED IN U.S.A.


GENERAL ELECTRIC
36C762855AB
CONT ON SHEET 3
SH NO 2
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE ELEMENTARY DIAGRAM
REVERSE LOGIC
FIRST MADE FOR STANDARD LINE
193X185AA601
CURRENT SIGNAL A
27
CURRENT SIGNAL C
28
CURRENT SIGNAL B
29
R487
47.5K
R488
47.5K
R489
47.5K
T482
2N4360
T483
2N4360
T484
2N4360
R522
22K
R523
22K
R524
22K
A. COM
+5V
+20V
R490
3.3K
R493
10K
IFFX
25
R491
10K
T477
2N4424
D477
R492
10K
R494
10K
R495
1MEG
A. COM
R496
3.3K
R499
10K
IFFC
A
R497
10K
T478
2N4424
D478
R498
10K
R500
10K
R501
1 MEG.
A. COM
FROM SHEET 1
R502
3.3K
R505
10K
IFFB
B
R503
10K
T479
2N4424
D479
R504
10K
R506
10K
R507
1 MEG.
A. COM
C478
.01MF
R509
82.5K
A58
IC5
R508
22K
A. COM
+20V
R483
10K
D476
T476
2N3417
CURRENT LIMIT ADJUST
22
R485
100Ω
R484
4.7K
-20V
P475
SLIP COMP
10K
P476
I. R. COMP
10K
I. R. COMP
20
A. COM
R510
100K
SLIP COMP
24
R521
100K
SLIP COMP
18
R519
10K
CURRENT LIMIT ADJUST
21
R511
1K
D480
T480
2N4424
D481
D483
D482
D484
D485
R525
470K
R526
470K
T481
2N5366
A. COM
D486
R518
10K
D487
R527
10K
A58
IC5
CURRENT LIMIT
26
R520
4.7K
A. COM
R512
22K
R513
10K
R514
3.3K
R515
22K
R516
3.3K
R517
10K
-20V
ICL
23
R545
4.7K
REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
JAN. 18, 1979
CHG. TO SH 3
SV-8174-3E
PRINTS TO
5B(8)M
5J(BW)
5R(2)BW
TD(1)
5L(2)BW
TS(50)
AW(BW)
5D(CD)
JA(CD)
MADE BY C.R. Baker MARCH 7, 1974
ISSUED E. Novak May 2, 1974
APPROVALS
SPEED VARIATOR
ERIE, PA.
DIV OR DEPT
LOCATION
36C762855AB
CONT ON SHEET 3
SH NO. 2
DRAWING NO.
REV NO.
FY-809-F (2-69)
PRINTED IN U.S.A.

GENERAL ELECTRIC

36C762855AB CONT ON SHEET FL SH NO 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ±	DECIMALS ±	ANGLES ±
716A515	✓			

TITLE PRINTED CIRCUIT DIAGRAM
REVERSE LOGIC

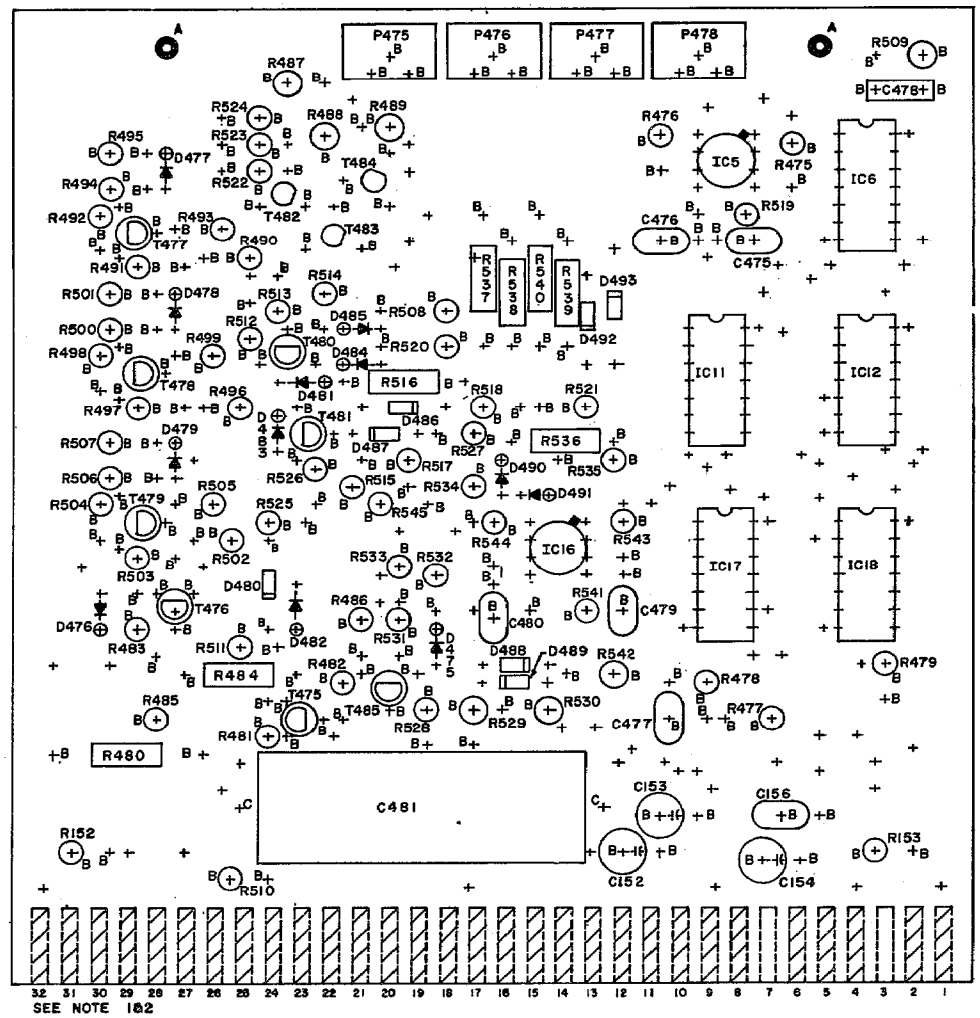
FIRST MADE FOR STANDARD LINE

193X185AAG01

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

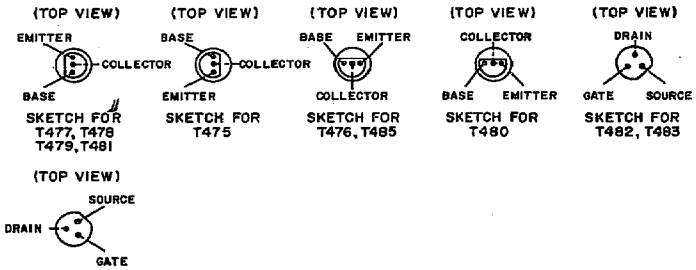
LOC.	DIA.	QUAN.
A	.157	2
B	.040	213
C	.052	2



NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES



2X	36A352970AB005	36A352971AB005	2	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	D. Baker JULY 5, 1974	CHG. PER ECO SV-874-32	
2	D. Baker MAR. 14, 1979	CHG. PER ECO SV-874-520	
3	D. Novick MAR. 16, 1979		

PRINTS TO

58(8)M	5J(BW)
5R(2)BW	TD(1)
5L(2)BW	TS(2)
2W(BW)	
5D(CD)	
1A(CD)	

MADE BY E.R. Baker MARCH 8, 1974
ISSUED L. Novick MAY 7, 1974

APPROVALS M.G. EAS

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

36C762855AB CONT ON SHEET FL SH NO 3

SCHEMATIC DIAGRAM

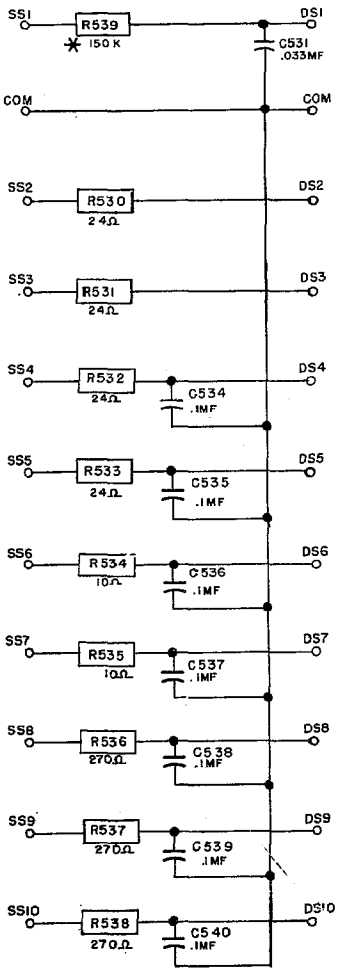
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	FRACTION	DECIMAL	ANGLE
718A515	✓	±	±	±

GENERAL ELECTRIC

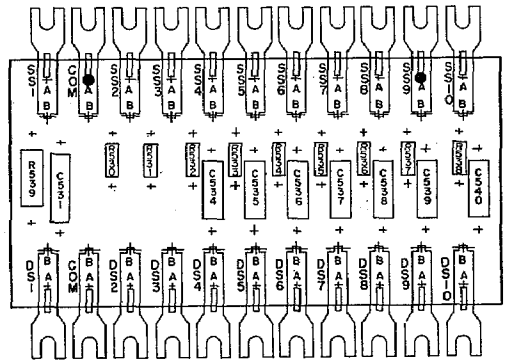
36C762868AA

PRINTED CIRCUIT DIAGRAM
SIGNAL FILTER
STANDARD LINE
193X189AAG01



✱ — INDICATES 1% RESISTANCES TOLERANCE

FRONT VIEW SHOWING LOCATION OF COMPONENTS



HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	22
B	.101	22

NOTE: 1. CARD SIZE, $2.85^{\pm.020}$ X $5.130^{+.002}_{-.008}$

36C762868AA

Ø 1

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT	
		FRONT	BACK
FULL	9428522AA004		

REVISIONS

1	J. Schmitz 1-03 Nov 22, 1971	KWP	50C(BK)55	5(S) 5D(BK) 5(PCM) 50 5B(2) 5(TS)50 5(QC)B

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

APPROVALS

J. Schmitz June 2, 1970 EAS
DCM JAN 5, 1971 KWP

SPEED VARIATOR
ERIE, PA.

36C762868AA

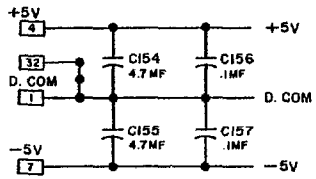
SCHEMATIC DIAGRAM

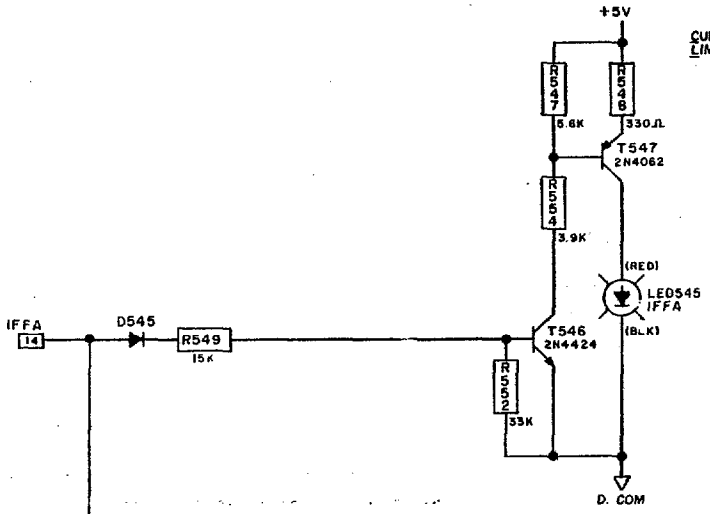
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:–				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
716A516	✓	±	±	±

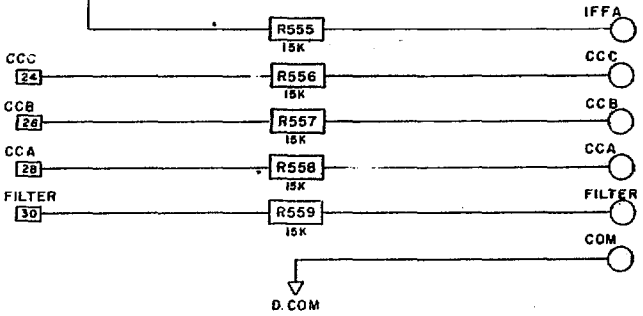
GENERAL ELECTRIC

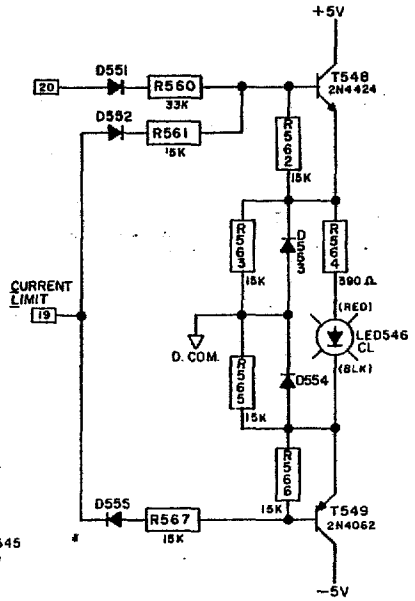
36C762854AA

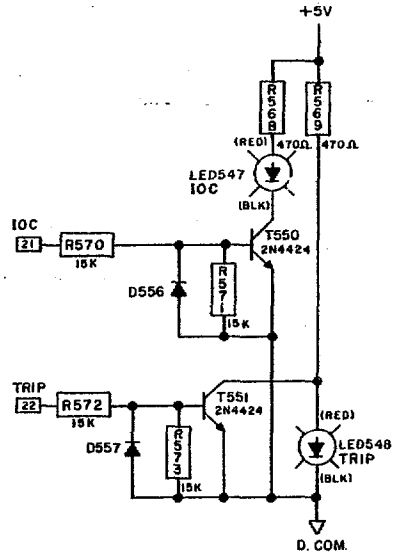
TITLE PRINTED CIRCUIT DIAGRAM
MONITOR
FIRST MADE FOR STANDARD LINE
193X184AAG01





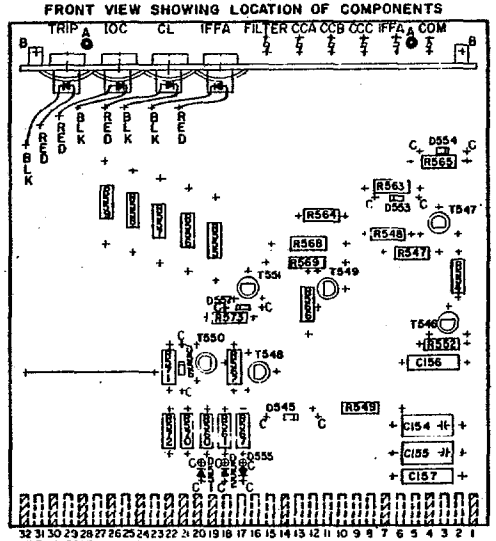






– INDICATES TEST CONNECTOR

FRONT VIEW SHOWING LOCATION OF COMPONENTS



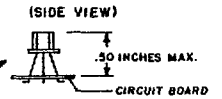
SEE NOTE 1 & 2

HOLE TABULATION

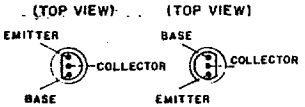
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

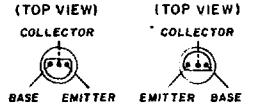
LOC.	DIA.	QUAN.
A	.157	2
B	.078	2
C	.032	16

NOTE 1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATE TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$
4. ALL TRANSISTORS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN HERE



TRANSISTOR LEAD SKETCHES





36C762854AA
REV NO 0

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	9428526AA004	1	0

REVISIONS

PRINTS

5QC(BK)5	5 (S)
	5D(BK)
	5B(2)
	5(TS)50
	5(QC)B

MADE BY	APPROVALS		
J. Britton JUNE 3, 1970	GAS	SPEED VARIATOR	36C762854AA
ISSUED K. Turner FEB 15 1971	KAP	ERIE, PENNA.	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC 36C762869BR

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
ISOLATOR / PS

FIRST MADE FOR STANDARD LINE
193X190AG601, 602

CONT ON SHEET 2 SH NO 1

NOTE:
1. RESISTORS MARKED WITH ** ARE 1% TOLERANCE (50 PPM / °C).
2. RESISTORS MARKED WITH * ARE 1% TOLERANCE (100 PPM / °C).
3. RESISTORS MARKED + ARE 1% TOLERANCE (100PPM / °C) AND ARE SELECTED IN TEST.

TABLE I

GROUP NUMBER	RESISTOR NOMENCLATURE AND VALUE	
	R593	R596
G01	182K **	182K **
G02	1K **	1K **

+20V INPUT 31 — R152 10Ω — R585 10Ω — +20V
R662 270Ω — TO POS. SUPPLY OF OPERATIONAL AMPLIFIER
R664 270Ω
D601 — C152 4.7MF — C585 .1MF — C595 .1MF — 15
C153 4.7MF — C586 .1MF
D602
R665 270Ω — C597 .1MF
R663 270Ω — TO NEG. SUPPLY OF OPERATIONAL AMPLIFIER
−20V INPUT 2 — R586 1K — −20V
+5V 4 — +5V
C154 4.7MF — C156 .1MF
1 — 32 COM — COM
IRX 6

+20V — R592 330Ω — D585 — RX
+5V — R589 2.7K — R590 10K — T585 2N4424 — R591 39K
26 RXNO
23 RXNO
25 RXNC
24 RXNC

DC LINK 9 — R593 ** SEE TABLE I — R595 2.74K — R600 150K — X — R601 ± — Y
C599 .01MF — R598 ** 22.1K
D593 — D594 — A58 OA152 — R597 ** 2.74K — R599 3.9K — R594 4.7K
D595 — D596 — A58 OA152 — R602 * 2.74K
VOLTAGE FDBK 5
+5V — R624 4.7K — I VOLTAGE TRIP 13 — T591 2N4424
D592 — R622 1K
DC MIDPOINT 11 — R596 ** SEE TABLE I
V — R603 * 12.1K — R607 4.7K — R608 * 5.62K — R610 4.7K — R618 100K — R623 6.8K
W — R604 * 22.1K — A58 OA153
+20V — R617 4.7K — −20V
+5V — T587 2N4424 — R611 2.7K — R613 4.7K — T588 2N4424
R620 100K — A41 OA154 — VOLTAGE FDBK FILTERED 14
R619 1K — C587 .47MF

26VDC 28 — D599 — R660 4.7K
25VAC 30 — D588 — R628 10Ω, 20W
25VAC 29 — D589 — R629 10Ω, 20W
D590 — D591 — R627 50Ω, 10W — R625 2.2K
R614 4.7K — T589 2N4424 — D COM
R605 10K — R609 22.1K — R612 6.8K — D586
R615 39.2K — R616 100K
T590 2N3417 — T594 TIP33A — R606 1K — D587
DELAYED FIRING POWER 18 19 20 21
Z585 — Z586 — Z587 — Z588 — R670 1K 2W — R666 1K — T592 2N4424 — R626 8.2K
+20V — R654 182K — +5V — R656 4.7K — C593 10MF — T598 2N4424 — R655 221K — R657 100K
T596 2N4424 — T597 D42C7 — R658 10Ω — C590 500MF — F COM 16 17
C592 .47MF — C594 10MF — D597 — −20V

D COM — R634 100Ω — R668 120Ω
R630 4.7K — R631 10K — A58 OA153 — R632 10K — T595 2N5366
R633 10K — −5V 7
C591 4.7MF — R635 330Ω — R669 750Ω — −20V

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS	PRINTS TO
	5(TS)50 5 (S)
	IT(BW) 5D(BK)
	JA(CD) 58(2)22
	5L(BW)5
	5QC(B)
	5QC(BK)5

MADE J.R. Baker MAR. 13, 1975
CHKD Paul Novak APR. 25, 1975
APVD
SPEED VARIATOR ERIE, PA.
DIV OR DEPT
LOCATION
36C762869BR
CONT ON SHEET 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FF-608-F (2-68) PRINTED IN U.S.A.

GENERAL ELECTRIC

36C762869BR
CONT ON SHEET FL SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

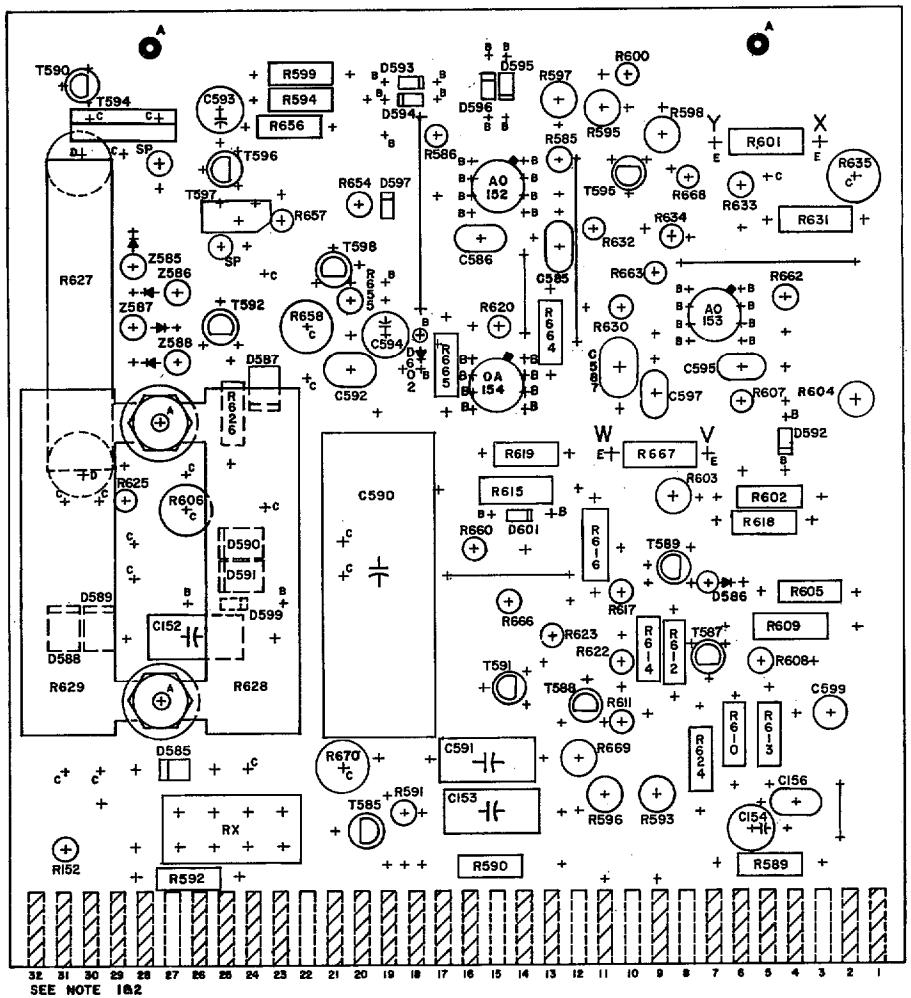
TITLE PRINTED CIRCUIT DIAGRAM
ISOLATOR / PS

FIRST MADE FOR STANDARD LINE
193X190AGG01,G02

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

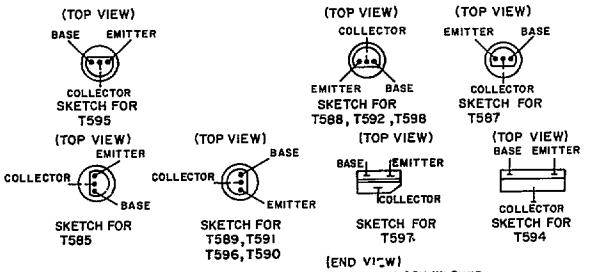
LOC.	DIA.	QUAN.
A	.157	4
B	.032	42
C	.062	21
D	.052	2
E	.078	4

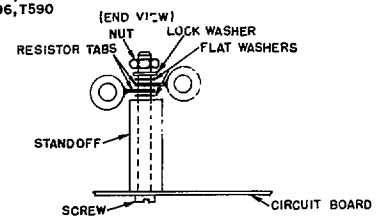


NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES





SKETCH FOR MOUNTING R628, R629

2X	36A352969BD004		2	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

5QC(B)	5 (S)
5QC(BK) 5	5D(BK)
	6B(2)22
	5T3(60

REV NO. 0

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY	APPROVALS	DIV OR DEPT	
J.R. Baker MAR. 13, 1975	EDS	SPEED VARIATOR	36C762869BR
ISSUED Wm Novak APR. 25, 1975	GHB	ERIE	CONT ON SHEET FL SH NO 2

GENERAL ELECTRIC

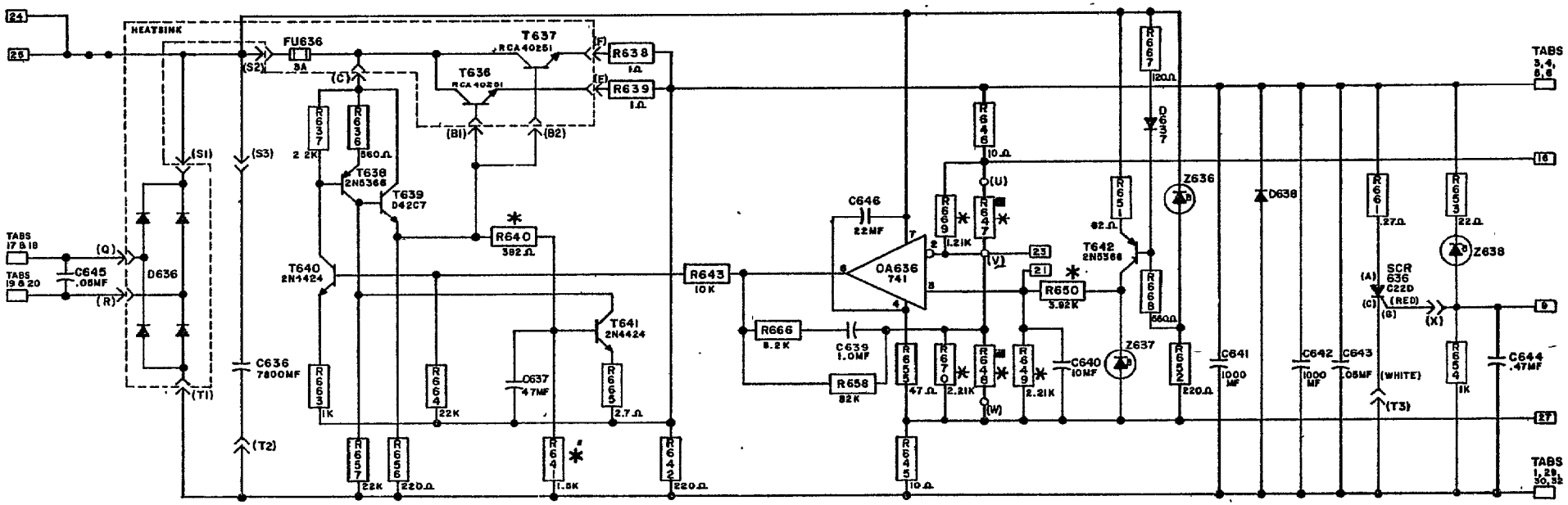
36C762870AB

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
+5 VOLT POWER SUPPLY

FIRST MADE FOR STANDARD LINE
193X191ABG01



✱ INDICATE 1% RESISTOR

■ RESISTORS SELECTED IN TEST

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REV NO
0

PRINTS TO

5QC(BK) 5	5(S)
5T(BW)	5D(BK)
5A(CD)	5B(2)22
5AE(BW)	5L(BW)5
	5(TS)50
	5(QC)8

MADE BY S. Mazza AUG 25, 1973

APPROVALS

SPEED VARIATOR ERIE, PA.

DIV OR DEPT LOCATION 36C762870AB-

CONT ON SHEET 2 SH NO. 1

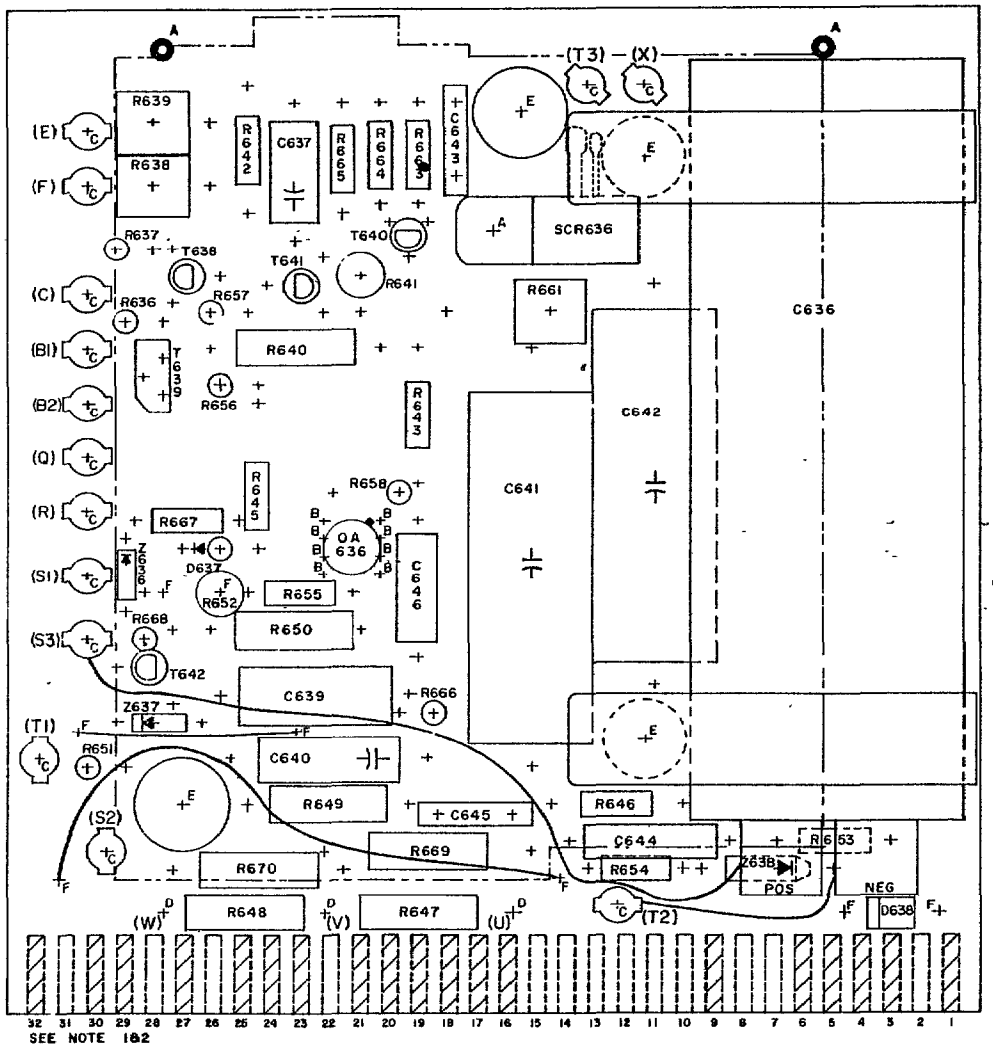
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C762870AB
CONT ON SHEET FL SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A815	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
+5 VOLT POWER SUPPLY
FIRST MADE FOR STANDARD LINE
193X191AB G01



HOLE TABULATION

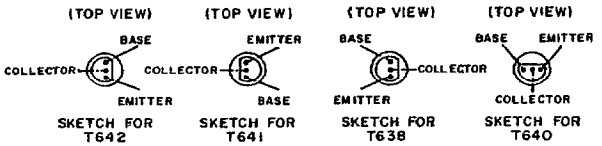
ALL HOLES 040 DIA. EXCEPT THE HOLES TABULATED BELOW

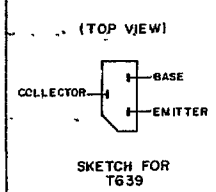
LOC.	DIA.	QUAN
A	.157	3
B	.032	8
C	101	14
D	078	3
E	.187	4
F	.062	8

NOTES

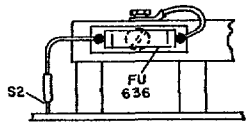
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.010}$ X $5.130^{+.002}_{-.008}$

TRANSISTOR LEAD SKETCHES

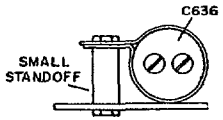




FUSE HOLDER CONNECTION SKETCH
(SIDE VIEW OF HEATSINK FROM TAB END OF CARD)



CAPACITOR MOUNTING SKETCH



LEAD CONNECTIONS

CONNECT RING TERMINAL OF LEAD S3 TO POSITIVE TERMINAL OF C636
CONNECT RING TERMINAL OF LEAD T2 TO NEGATIVE TERMINAL OF C636
COUPLE RED LEAD FEMALE CONNECTOR TO MALE CONNECTOR X AND SCR636 WHITE LEAD FEMALE CONNECTOR TO MALE CONNECTOR T3
COUPLE REMAINING LEADS WITH FEMALE CONNECTORS TO MALE CONNECTORS BEARING NOMENCLATURE IDENTICAL TO THE LEAD MARKING

2X	36A352968AB004	3	0
SCALE	ETCHED CIRCUIT BOARD DWG.	FRONT SPACE	BACK UNIT REQUIREMENT

REVISIONS

PRINTS TO	
5L(BW)5	5 (S)
5T(BW)	5D(BK)
5JA(CD)	5B(2)22
5AE(BW)	5T8(5D
	5QC(BK)
	5QC(B)

MADE BY	SEPT 4,1973	APPROVALS	SPEED VARIATOR	36C762870AB
ISSUED	FEB 1 1974		ERIE, PA	CONT ON SHEET FL SH NO 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

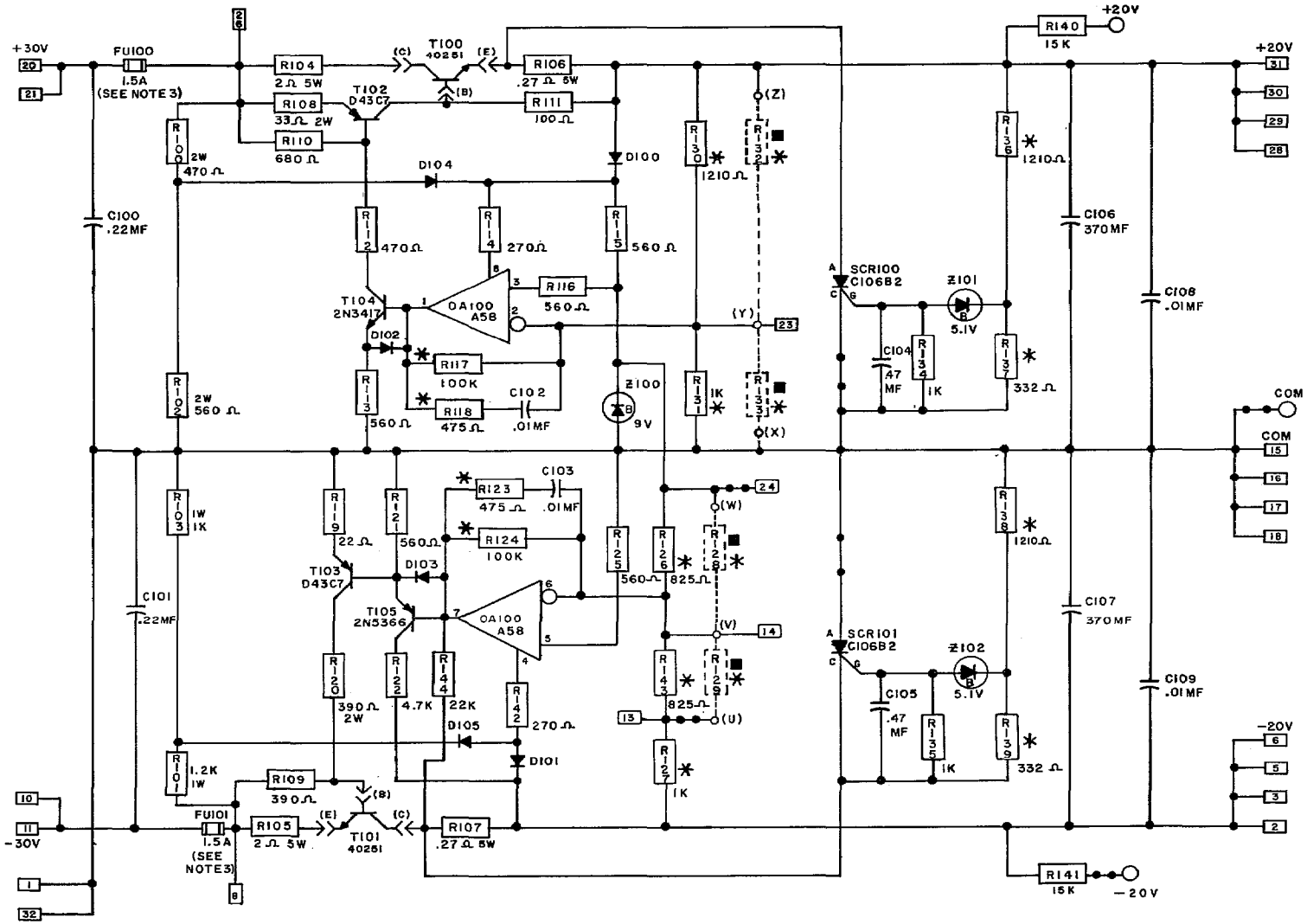
GENERAL ELECTRIC

36C762896AB
CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A505	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
20 VOLT POWER SUPPLY CARD
FIRST MADE FOR STANDARD LINE
193X257AAG01



NOTES 1. ✱ — 1% TEMPERATURE COMPENSATED RESISTOR.
■ — RESISTOR SELECTION IN TEST.

2. NUMBERS INSIDE SMALL RECTANGLES INDICATE TAB NUMBERS WHICH CORRESPOND TO MATCHING RECEPTACLE NUMBERS.

3. FUSES ARE INSTRUMENT TYPE, LITTLE FUSE 361001, OR EQUIVALENT.

REV NO. 0

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

PRINTS TO

JA(CD)	5(S)
IT(BW)	5QC(BK) 5
5L(BW)5	5D(BK)
5AE(BW)	5B(2)22
	5(TS)50
	5(QC)BE

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY C.R. Baker MARCH 15, 1974
ISSUED G. Mowat SEPT. 4, 1974
APPROVALS
speed variator DIV. OR DEPT.
ERIE, PA. LOCATION

36C762896AB
CONT ON SHEET 2 SH NO 1

GENERAL ELECTRIC

36C762896AB FL 2

PRINTED CIRCUIT DIAGRAM
20 VOLT POWER SUPPLY

FIRST MADE FOR STANDARD LINE
193X257AA601

UNLESS OTHERWISE SPECIFIED, TOLERANCES IN MACHINE DIMENSIONS ARE AS FOLLOWS:

APPLIED PRACTICES	SURFACES	TOLERANCES IN MACHINE DIMENSIONS (TWO PLACE / FOUR PLACE / ANGULAR)
718A515	√	± / ± / ±

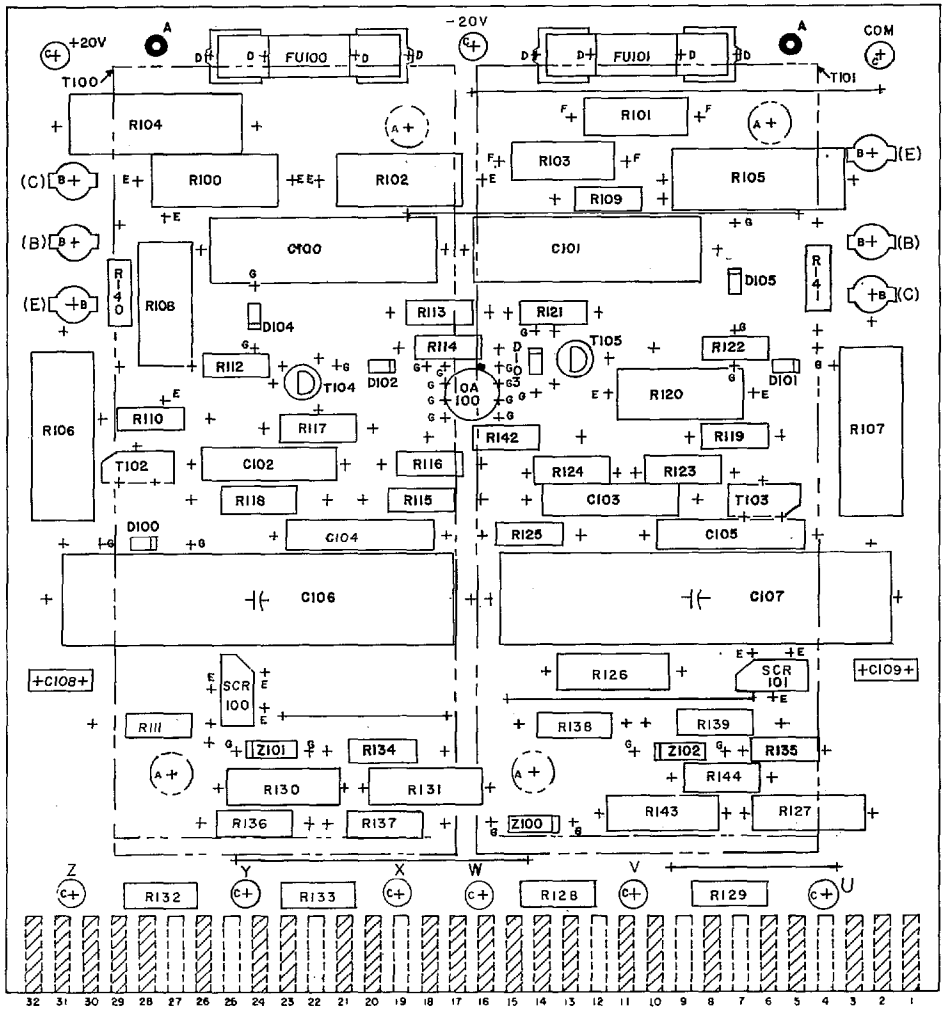
HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	6
B	.101	6
C	.081	9
D	.078	8
E	.062	14
F	.052	4
G	.032	26

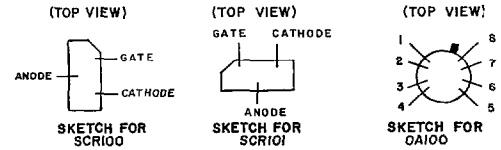
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.010}$ X $5.130^{+.002}_{-.008}$

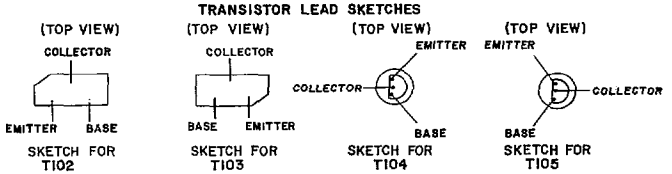


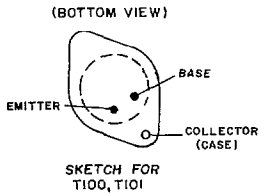
GROUP	KEY LOCATIONS			
01	6-7	12-13	26-27	—

SCR LEAD SKETCHES



TRANSISTOR LEAD SKETCHES





2X	36A353740AB004		2	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PHS. S.T.	
JA(CD)	5 (S)
IT(BW)	5D(BK)
5L(BW)	5B(2)22
5AE(BW)	5QC(BK)5
	5(TS)50
	5QC(B)

MADE BY G.H.Baker MARCH 15, 1974
APPROVED R.Novak SEPT. 4, 1974

speed variator
ERIE, PA.

36C762896AB FL 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

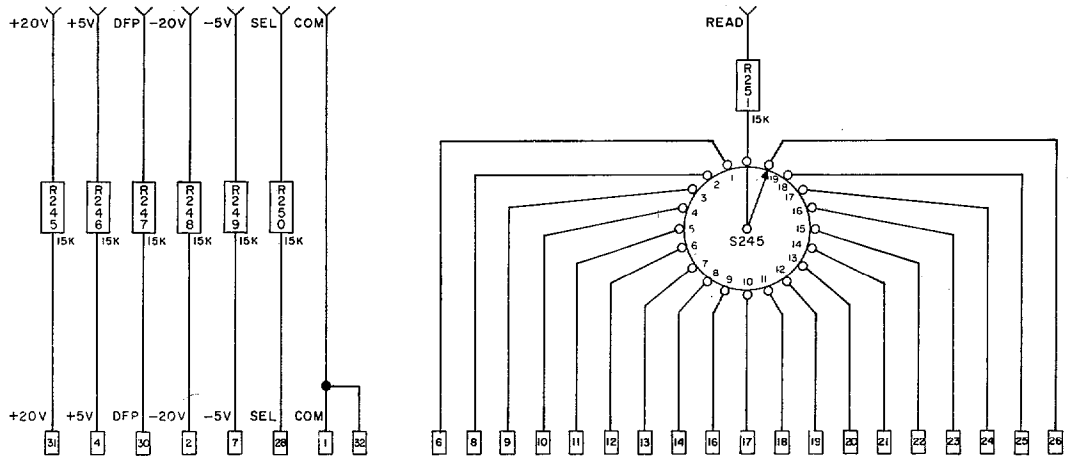
GENERAL ELECTRIC | 36C762858AA

SCHEMATIC DIAGRAM

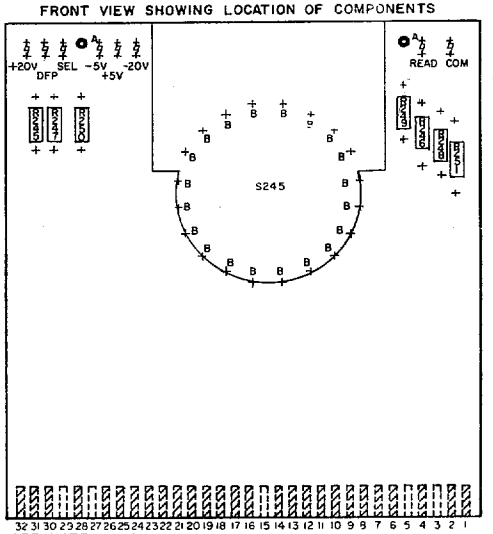
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
TEST CARD

FIRST MADE FOR STANDARD LINE
193X188AAG01



FRONT VIEW SHOWING LOCATION OF COMPONENTS



SEE NOTE 1 & 2

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.062	20

NOTE 1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.

2. CROSS HATCHED TABS INDICATE TABS USED.

3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

36C762858AA REV 0

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	9428549AA004	2	0

REVISIONS

PRINTS TO
5QC(BK05 5B(S)
5D(BK
5B(2
5(TS)5(
5(QC)

MADE BY G.R. Baker NOV. 24, 1970 APPROVALS EAS KWP

speed variator ERIE, PA.

DIV OR DEPT LOCATION 36C762858AA CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764129AA

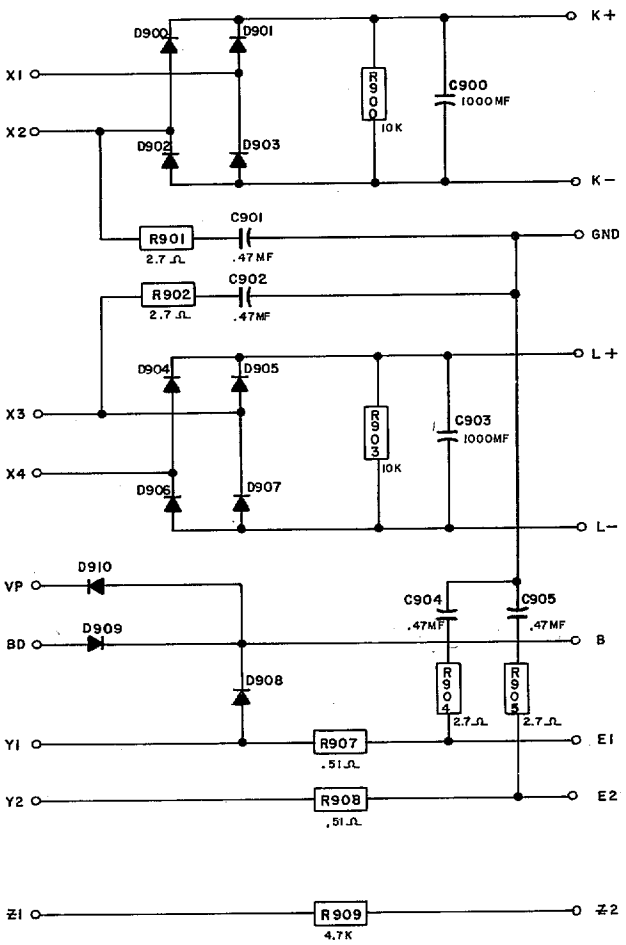
CONT ON SHEET | SH NO.

SCHEMATIC DIAGRAM

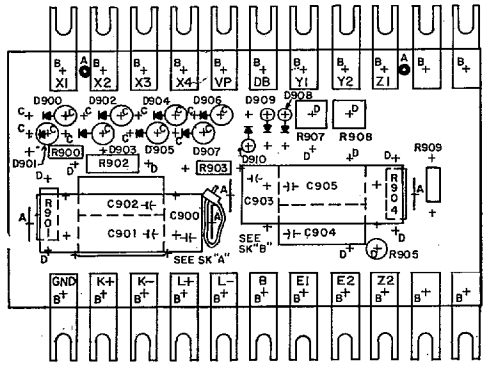
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
POWER INTERFACE
FIRST MADE FOR STANDARD LINE
193X370AA G01



FRONT VIEW SHOWING LOCATION OF COMPONENTS

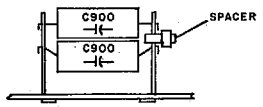


HOLE TABULATION

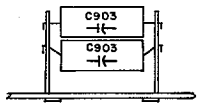
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	6
B	.101	22
C	.062	16
D	.052	12

CARD SIZE, $2.85 \pm .020$ X $5.130^{+.002}_{-.008}$



SKETCH "A"



SKETCH "B"

REV NO. 0

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	36B605049AA004		

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

5QC(BK)5	5 (S)
IT(BW)	5D(BK)
	5(TS)50
	5B(2)22
	JA(CD)
	5(QC)B

MADE BY W. Gabo JUNE 20, 1973
ISSUED C. R. Baker by FEB. 1, 1974

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

36C764129AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764140AA

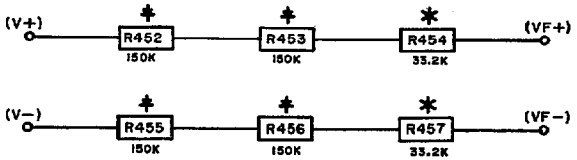
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

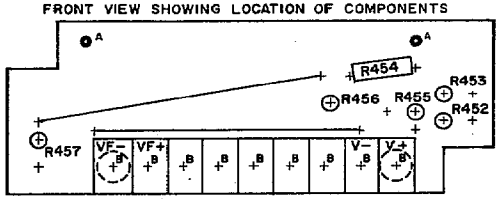
TITLE PRINTED CIRCUIT DIAGRAM
RESISTANCE ISOLATOR CARD

FIRST MADE FOR STANDARD LINE
193X263AAG03

SCHEMATIC DIAGRAM



FRONT VIEW SHOWING LOCATION OF COMPONENTS



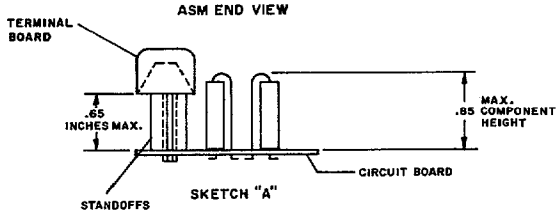
HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.078	9

NOTE—1. CARD SIZE, $1.90^{+.000}_{-.015}$ X $5.130^{+.002}_{-.006}$

ASM END VIEW



SKETCH "A"

NOTES

1. ✱ 1% RESISTOR — 50 PPM
2. ✚ .2% RESISTOR — 50 PPM

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMEN FRONT	BACK
FULL	36B605027AA004		

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varrator Department

PRINT

5QC(BK)5	JA(C
5B(2)22	5D(B
5(TS)50	1T(B
5(S)	5L(B
	5AE(B
	5(QC

DRAWN BY J.R. Baker DEC. 17, 1973
ISSUED Earl Novak MARCH 26, 1974
APPROVALS GBS JJB
speed varidor ERIE, PA.
DIV OR DEPT LOCATION
36C764140AA CONT ON SHEET DH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

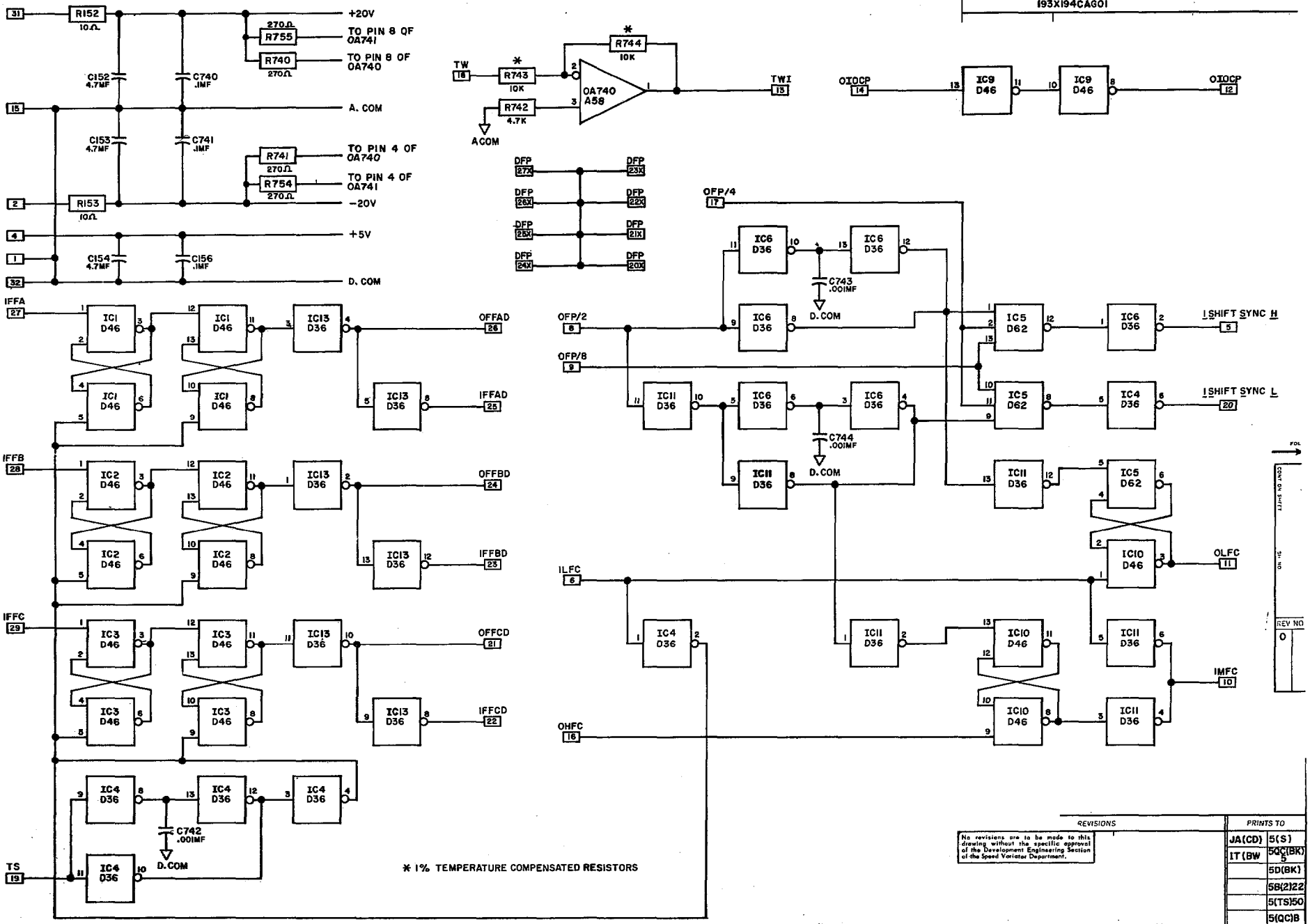
36C762893CA
CONT ON SHEET 2 SH NO. 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
PHASE SHIFT

FIRST MADE FOR STANDARD LINE
193X194CAG01



REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

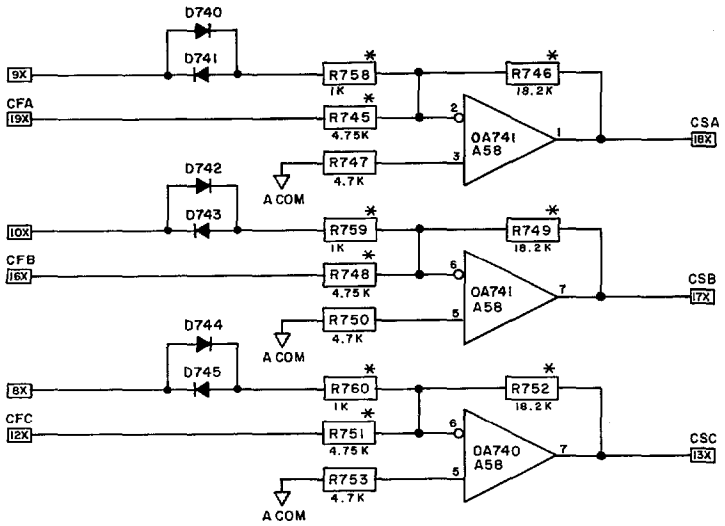
JA(CD)	5(S)
IT(BW	5QC(BK) 8
	5D(BK)
	5B(2)22
	5(TS)50
	5(QC)8

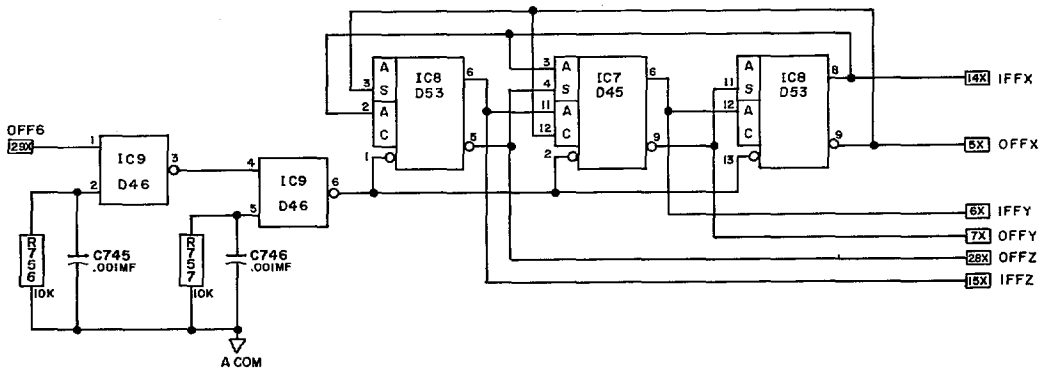
MADE BY G. Martin AUG. 31, 1972
ISSUED D. Turner MARCH 15, 1974
APPROVALS

speed variator DIV OR DEPT
ERIE, PA. LOCATION

36C762893CA
CONT ON SHEET 2 SH NO. 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY





PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

JA(CD)	5B(2)22
IT (BW)	5D(BK)
	5(QC)8
	5QC(BK)5
	5(TS)50
	5(S)

GENERAL ELECTRIC

36C762893CA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES: 710A818 | SURFACES: ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

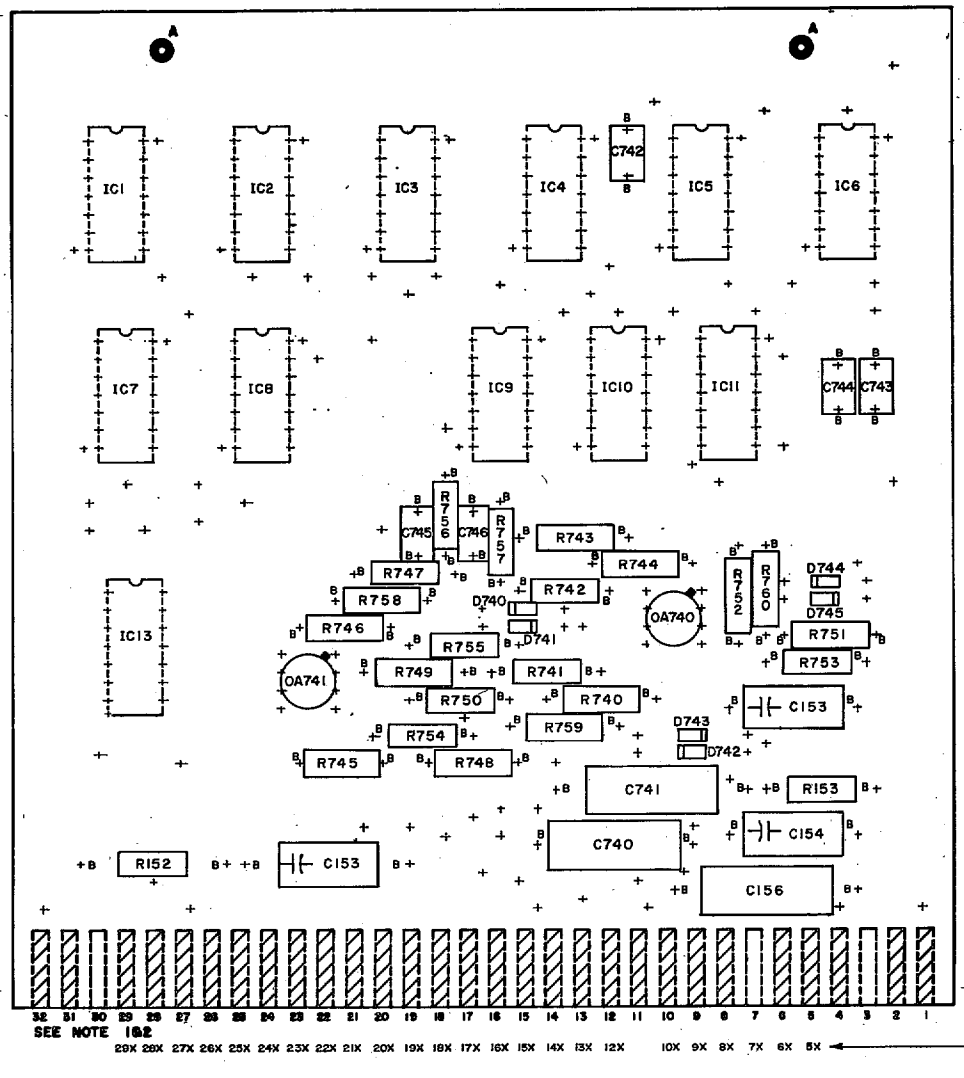
TITLE PRINTED CIRCUIT DIAGRAM
PHASE SHIFT CARD
FIRST MADE FOR STANDARD LINE
193X194CA601

CONT ON SHEET FL. SH NO 3

HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

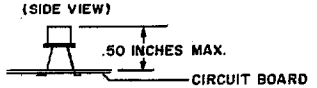
LOC.	DIA.	QUAN.
A	.157	2
B	.040	68



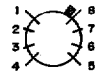
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.150^{+.002}_{-.008}$
4. THIS CARD HAS GOLD PLATED TABS ON BOTH SIDES. TABS I THRU 32 ARE LOCATED ON THE REVERSE SIDE. TABS IX THRU 32X ARE LOCATED ON THE COMPONENT SIDE OF THE CARD. TAB IX IS OPPOSITE TAB I AND ETC. THE TAB NUMBERS SHOWN ARE THOSE USED ON THIS CARD.
5. ALL OP AMPS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

OP AMP (SIDE VIEW)

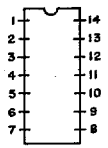


OP AMP LEAD SKETCH (TOP VIEW)



SKETCH FOR OA740, OA741

IC LEAD SKETCH (TOP VIEW)



SKETCH FOR IC1, IC2, IC3, IC4, IC5, IC6, IC7, IC8, IC9, IC10, IC11, IC13

2X	36A352957AC005	36A352958AC005	1	0
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV NO. 0

PRINTS TO

1T(BW)	5B(2)22
JA(CD)	5D(BK)
	5(QC)B
	5QC(BK)5
	5(TS)50
	5(S)

DRAWN BY G. Mattei JUNE 11, 1973
ISSUED MARCH 15, 1974
APPROVALS
SPEED VARIATOR PROD DEV OR DEPT
ERIE, PA. USA LOCATION

36C762893CA
CONT ON SHEET FL. SH NO 3

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY